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Master Thesis

Computational algorithms for pricing financial derivatives under local volatility models

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SUMMARY

Abstract here

Keywords: Heston model; COS method; implied volatility; neural surrogate; calibration; physics-informed neural networks; Greeks.

DEDICATION

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1. INTRODUCTION

1.1. Motivation

Modern derivative engines must do more than price options: they must calibrate models repeatedly and produce stable sensitivities for risk management. In the Heston stochastic-volatility model [1], semi-analytical and Fourier-based methods provide a strong numerical foundation because the characteristic function is available. In particular, the Fourier-COS method gives accurate and efficient reference prices when the truncation and expansion settings are controlled [2].

The practical difficulty is that these routines are rarely called only once. Calibration requires repeated evaluations of the pricing map, large benchmark datasets require many reference valuations, and Greek analysis requires stable local derivatives. This repeated-evaluation setting makes neural approximators attractive, but speed alone is not sufficient in a financial workflow. A learned component is useful only if its errors are measurable, its failure regions are visible, and its outputs remain compatible with the downstream task.

This thesis studies neural methods from that perspective. The question is not whether neural networks can be fast, but whether they can be made reliable inside a benchmarked derivative workflow. The objective is therefore to identify where neural components can accelerate Heston pricing, calibration, and Greek computation while remaining anchored to explicit numerical references and clearly stated limitations.

1.2. Problem Setting

The thesis focuses on European vanilla options under the Heston model. This setting is narrow enough to admit reliable reference methods, but rich enough to expose the main difficulties of neural derivative analytics: nonlinear parameter effects, repeated inverse problems, low-price regions, short maturities, and non-smooth payoff behavior.

The main methodological risk is that average pricing accuracy can hide downstream failures. A surrogate with small mean pricing error may still distort calibration if its residuals are structured across moneyness or maturity. A differentiable pricing network may produce plausible prices while giving unstable Delta or Gamma. These issues are especially relevant near the short-maturity at-the-money region, where the payoff kink creates a structural challenge for second-order sensitivities.

For this reason, the thesis treats benchmarking as part of the method rather than as an afterthought. Numerical prices, implied-volatility labels, calibrated parameters, PINN prices, and Greeks are all evaluated under explicit reference conventions. This makes it possible to separate acceleration from reliability, and to distinguish useful neural compo-

nents from model variants that only improve a single metric in isolation.

1.3. Methodological Approach

The workflow combines a numerical reference lane with a neural approximation lane. The reference lane produces controlled Heston-COS prices, implied-volatility labels, and semi-analytical Greek benchmarks. The neural lane uses COS/IV labels for the ANN surrogate, embeds the surrogate in a CaNN calibration loop, and evaluates PINN pricing and Greek components against the reference objects. The base PINN is trained through the PDE formulation; reference Greeks enter only through benchmark diagnostics and the Sobolev-enhanced variant.

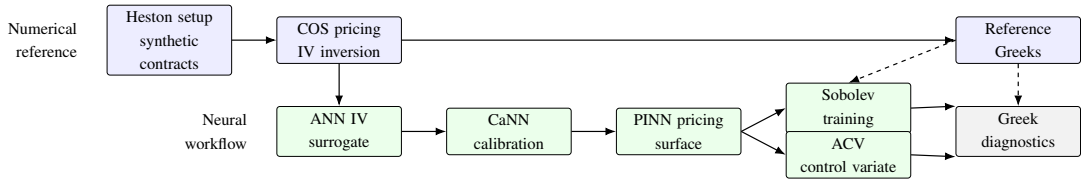


Fig. 1.1. High-level structure of the benchmarked Heston workflow. Solid arrows show the main computational flow; dashed arrows show reference Greeks used to compute diagnostic errors and to define Sobolev Greek targets.

This organization is important for the interpretation of the results. The ANN is judged not only as a fast function approximator, but also as the forward engine used by the calibration routine. The PINN is judged not only by price RMSE, but also by spatial error structure and derivative quality. The Greek analysis is treated as a separate problem because automatic differentiation through a learned price surface does not by itself guarantee reliable sensitivities.

1.4. Main Contributions

The first contribution is a reproducible benchmark pipeline for Heston derivative analytics. The pipeline connects data generation, COS reference pricing, implied-volatility inversion, ANN training, CaNN calibration, PINN pricing, and Greek diagnostics under common conventions.

The second contribution is the improved neural calibration result. Building on the Liu-style calibration framework [3], the final $\tanh$ ANN/CaNN configuration improves the controlled Liu et al. benchmark in the reported calibration setting, especially in parameter recovery.

The third contribution is a PINN pricing and sensitivity benchmark under Heston. The PINN is evaluated against COS and semi-analytical references not only at the price level, but also through first- and second-order Greeks.

The fourth contribution is the analysis of Greek improvement methods in the hard short-maturity at-the-money region. Sobolev-style derivative supervision [4] and the analytic-control-variate diagnostic improve Greek quality, while the remaining errors show that the payoff kink remains a structural limitation.

1.5. Scope and Thesis Structure

The scope is intentionally controlled. The experiments use European vanilla options under the Heston model, synthetic data with known reference values, and benchmark protocols where numerical settings can be inspected. Market microstructure effects, bid-ask spreads, dividends, exotic payoffs, and portfolio-scale deployment are outside the main experimental scope. This restriction is deliberate: it allows the thesis to test the numerical and neural components before making broader claims about market deployment.

The remainder of the thesis is organized as follows. Chapter 2 introduces the mathematical and financial background, including the Heston model, characteristic-function pricing, the COS method, semi-analytical Greeks, and implied-volatility inversion. Chapter 3 presents the supervised ANN implied-volatility surrogate. Chapter 4 describes the CaNN calibration workflow. Chapter 5 develops the PINN pricing and Greek computation framework. Chapter 6 reports the benchmark results for ANN pricing, CaNN calibration, PINN pricing, Sobolev Greeks, and the ACV diagnostic. Chapter 7 summarizes the conclusions, limitations, and future work.

2. MATHEMATICAL AND FINANCIAL BACKGROUND

2.1. State Of The Art

The literature relevant to this thesis can be grouped into three blocks. First, classical model-based pricing combines closed or semi-closed formulas (e.g., Black–Scholes and Heston) with deterministic numerical solvers under explicit stochastic assumptions [1], [5]. These methods are financially interpretable and robust, but repeated calls in calibration or surface-generation loops can be computationally expensive.

Second, spectral/Fourier pricing methods exploit characteristic functions to obtain fast and accurate valuation formulas, with FFT- and COS-based implementations as standard references [2], [6]. Their main strength is the accuracy-speed tradeoff in repeated pricing, while practical performance still depends on stable parameterization choices (truncation interval, series size, and numerical inversion settings).

Third, neural surrogates learn the pricing map directly from synthetic or market-consistent labels. Feed-forward ANN surrogates have shown substantial acceleration in calibration workflows [3]. More recently, PINN-type approaches have been proposed to inject structural constraints and improve data efficiency [7], [8]. However, the PINN literature also reports optimization pathologies, sensitivity to loss balancing, and robustness issues outside the training distribution [9], [10], [11].

Within this landscape, this thesis reuses the Heston-COS framework as high-accuracy numerical reference and contributes an empirical comparison of data-driven alternatives (ANN and PINN-family variants) under a common data-generation, evaluation, and benchmarking protocol.

TABLE 2.1. METHODOLOGICAL BLOCKS AND ROLE IN THIS
THESIS

Block	Representative methods	Main strengths	Main limitations
Classical model-based pricing	Black–Scholes, Heston semi-closed pricing [1], [5]	Financial interpretability; strong theoretical grounding	High computational cost in repeated evaluations
Spectral pricing	FFT/Carr–Madan, COS/Fang–Oosterlee [2], [6]	High accuracy with efficient repeated pricing	Requires careful numerical setup ($[a, b]$, N , inversion stability)
Data-driven surrogates	ANN pricers/calibrators [3]	Very fast inference once trained	Domain-shift and extrapolation risk
Physics-informed surrogates	PINN/gPINN variants [7], [8]	Injects structural constraints into learning	Training instability and sensitive optimization [9], [10], [11]

Source: Author’s synthesis from the cited literature

2.2. Mathematical Framework For Option Pricing

This section defines the common mathematical setup used in the rest of the thesis for pricing European-style derivatives.

Time is continuous on $[0, T]$, with current time t and maturity T . A derivative is a financial contract whose value depends on an underlying asset price process S_t ; in particular, S_T denotes the underlying price at maturity. A European option gives its holder a right, but not an obligation, exercisable only at maturity T : a call option pays when S_T finishes above the strike K , while a put option pays when S_T finishes below K . For example, the terminal payoff is $(S_T - K)^+$ for a European call and $(K - S_T)^+$ for a European put, where $(x)^+ = \max(x, 0)$. Rather than introducing the full probabilistic machinery, we keep a pricing-focused notation that is sufficient for the methods used later in the thesis.

A global notation glossary is provided in Appendix D and will be extended as new symbols are introduced.

As a practical starting point for this thesis, we use the standard risk-neutral valuation formula for European claims with constant rate r [12, Ch. 8]:

$$V_t = e^{-r\tau} \mathbb{E}_t^{\mathbb{Q}}[\Psi]. \quad (2.1)$$

Here, V_t is the option value at time t , $\tau = T - t$ is time to maturity, Ψ is the terminal

payoff, and $\mathbb{E}_t^{\mathbb{Q}}[\cdot]$ denotes conditional expectation under the risk-neutral measure $\mathbb{Q}$. The risk-neutral measure is the pricing probability measure under which discounted tradable assets have no expected excess return; this is why future payoffs can be discounted at the risk-free rate r in this formula.

In the following sections, this framework is connected to three ingredients used throughout the thesis: the stochastic-calculus tools needed for the model equations, the Heston stochastic-volatility dynamics, and the characteristic-function/Fourier-COS pricing route.

2.3. Stochastic Calculus Background

This section introduces only the stochastic-calculus notions needed by the pricing methods used later. The presentation follows standard probability definitions and the transform-pricing notation adopted in [12, Ch. 1] and [2, Sec. 2].

For a real random variable X , the cumulative distribution function is

$$F_X(x) = \mathbb{P}(X \leq x). \quad (2.2)$$

Here, x is a real evaluation point and $\mathbb{P}$ denotes probability. If F_X is differentiable, the associated probability density function is

$$f_X(x) = \frac{\partial F_X(x)}{\partial x}. \quad (2.3)$$

Following this notation, the characteristic function of X is

$$\phi_X(u) := \mathbb{E}[e^{iuX}] = \int_{-\infty}^{\infty} e^{iux} dF_X(x) = \int_{-\infty}^{\infty} e^{iux} f_X(x) dx, \quad (2.4)$$

for $u \in \mathbb{R}$, where i is the imaginary unit.

Applying the logarithm to ϕ_X , we obtain the cumulant characteristic function

$$\zeta_X(u) = \log \mathbb{E}[e^{iuX}] = \log \phi_X(u). \quad (2.5)$$

Its Maclaurin expansion is

$$\zeta_X(u) = \sum_{k=0}^{\infty} \frac{\partial^k \zeta_X(u)}{\partial u^k} \Big|_{u=0} \frac{u^k}{k!} = \sum_{k=0}^{\infty} \zeta_k \frac{u^k}{k!}, \quad (2.6)$$

where $\zeta_k \equiv \frac{\partial^k \zeta_X(u)}{\partial u^k} \Big|_{u=0}$ is the k -th cumulant.

The moment generating function is

$$\mathcal{M}_X(u) := \phi_X(-iu) = \mathbb{E}[e^{uX}]. \quad (2.7)$$

Using $\mathcal{M}_X(u) = \phi_X(-iu)$ and $\zeta_X(u) = \log \mathcal{M}_X(u)$, cumulants can be computed from the characteristic function as

$$\zeta_k = \frac{\partial^k \zeta_X(u)}{\partial u^k} \Big|_{u=0} = \frac{\partial^k}{\partial u^k} \log \phi_X(-iu) \Big|_{u=0}. \quad (2.8)$$

These objects are central in the rest of this chapter: the characteristic function provides direct access to Fourier-domain pricing formulas, and low-order cumulants are later used to construct truncation intervals for the COS method.

2.4. The Heston Stochastic Volatility Model

In a general diffusion form, the Heston dynamics can be written as

$$dS_t = \mu S_t dt + \sqrt{\nu_t} S_t dW_t^{(S)}, \quad (2.9)$$

$$d\nu_t = \kappa(\bar{\nu} - \nu_t) dt + \gamma \sqrt{\nu_t} dW_t^{(\nu)}, \quad (2.10)$$

$$d\langle W^{(S)}, W^{(\nu)} \rangle_t = \rho dt. \quad (2.11)$$

Here, ν_t is the instantaneous variance, $W_t^{(S)}$ and $W_t^{(\nu)}$ are Brownian motions driving the price and variance shocks, and $d\langle W^{(S)}, W^{(\nu)} \rangle_t$ denotes their instantaneous quadratic co-variation. The parameter μ denotes a generic drift under the original measure. In the no-dividend risk-neutral pricing form used in this thesis, the drift becomes r :

$$dS_t = rS_t dt + \sqrt{\nu_t} S_t dW_t^{(S, \mathbb{Q})}. \quad (2.12)$$

This is the form used in the pricing formulas of the following sections [1], [12].

Here, ν_t is the instantaneous variance (equivalent to the ν_t notation used in later data-driven chapters). The variance process is of CIR type, so it is mean-reverting toward $\bar{\nu}$ at speed κ , with volatility of variance controlled by γ .

The five Heston parameters have distinct financial roles. The pair $(\bar{\nu}, \nu_0)$ controls the level of variance, with ν_0 mainly affecting the short end and $\bar{\nu}$ anchoring longer maturities. The speed κ determines how quickly variance reverts toward its long-run level. The parameter γ controls the variability of the variance process itself, and therefore the curvature of option smiles. Finally, ρ links spot and variance shocks; in equity-index applications it is usually negative, which generates the left-skewed implied-volatility patterns observed in markets.

This parameter interpretation is directly linked to the implied-volatility (IV) surface, i.e., the map of implied volatilities across strikes and maturities. At a fixed maturity, the cross-section of this surface is often called the volatility smile:

TABLE 2.2. INTERPRETATION OF HESTON PARAMETERS

Parameter	Interpretation	Main effect on IV surface
ρ	Instantaneous correlation between spot and variance shocks.	Controls skew: more negative ρ typically implies stronger left skew.
γ	Volatility of variance (<i>vol-of-vol</i>).	Controls curvature and short-maturity smile amplitude; larger γ usually increases convexity.
$\bar{\nu}$	Long-run mean level of the variance process.	Sets the global IV level around which the surface oscillates across maturities.
ν_0	Initial variance level at valuation time (ν_{t_0}).	Anchors short-maturity ATM IV and strongly affects the front end of the term structure.
κ	Speed of mean reversion of ν_t toward $\bar{\nu}$.	Controls how fast maturities converge to long-run variance, affecting term-structure smoothing.

Source: Based on [1] and [13, Chs. 2–3]

In this thesis, implied volatility is defined as the value σ_{imp} that reproduces a given option price when inserted into the Black–Scholes formula with the same contract and market inputs [5]:

$$V_{BS}(S_0, K, r, \tau, \sigma_{imp}) = V_{target}. \quad (2.13)$$

Here, S_0 is the initial value of the underlying price process, V_{BS} is the Black–Scholes price, and V_{target} is the target option value to be matched. Therefore, implied volatility is a quoting transformation of prices (not a structural parameter of Heston). It is nevertheless the natural comparison space for calibration and machine-learning experiments, because market option quotes are usually summarized across strikes and maturities through IV surfaces rather than raw prices. The numerical inversion used to compute σ_{imp} is detailed in Section 2.7.

A common positivity condition for the CIR variance process is the Feller condition

$$2\kappa\bar{\nu} \geq \gamma^2. \quad (2.14)$$

In practice, calibrated parameter sets may violate this inequality while still producing usable market fits; therefore, it is treated as a diagnostic and stability indicator rather than as a hard calibration constraint in many numerical workflows [13, Chs. 2–3].

For pricing, the key advantage of Heston is that the log-price characteristic function is available in closed (semi-analytical) form [1]. This makes the model especially suit-

able for Fourier-domain valuation, which is the computational route adopted in the next sections.

2.5. Pricing Via Characteristic Functions

When the characteristic function is available in closed or semi-closed form, pricing in the Fourier domain becomes a natural alternative to direct density integration. Following [2, Sec. 2], we write the option value in terms of the conditional transition density of the terminal state variable.

Using the notation adopted in the next section, with $x = X(t_0)$ and $y = X(T)$, the discounted valuation formula is

$$V(t_0, x) = e^{-r\tau} \int_{\mathbb{R}} V(T, y) f_X(T, y; t_0, x) dy. \quad (2.15)$$

Here, t_0 is the valuation date, X is the state variable used for transform pricing, $V(T, y)$ is the terminal payoff as a function of the terminal state y , and $f_X(T, y; t_0, x)$ is the conditional transition density of $X(T) = y$ given $X(t_0) = x$. The corresponding conditional characteristic function is

$$\phi_X(u; t_0, x) := \mathbb{E}_t^{\mathbb{Q}}[e^{iuX(T)} | X(t_0) = x] = \int_{\mathbb{R}} e^{iuy} f_X(T, y; t_0, x) dy. \quad (2.16)$$

Under standard integrability conditions, the density can be recovered by Fourier inversion:

$$f_X(T, y; t_0, x) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iuy} \phi_X(u; t_0, x) du. \quad (2.17)$$

Let $\hat{V}_T(u) := \int_{\mathbb{R}} e^{-iuy} V(T, y) dy$ denote the Fourier transform of the payoff at maturity. Substituting the inverse representation of f_X into the pricing integral and exchanging the order of integration gives

$$V(t_0, x) = e^{-r\tau} \frac{1}{2\pi} \int_{\mathbb{R}} \phi_X(u; t_0, x) \hat{V}_T(u) du. \quad (2.18)$$

Compared with PDE solvers, this approach avoids full space-time gridding. Compared with Monte Carlo, it is deterministic and does not introduce sampling noise. In repeated-pricing tasks (surface generation, calibration loops, and large synthetic datasets), this is a major practical advantage.

The Fourier-COS method used in the next section is a specific spectral discretization of this transform-based pricing framework: it replaces the continuous integral by a finite cosine expansion over a truncated interval, preserving high accuracy with efficient vectorized evaluation.

2.6. The Fourier-COS method

The Fourier-COS method approximates transition densities and option values by a cosine series on a finite interval [2, Secs. 3–4]. Its practical appeal in this thesis is that it combines spectral convergence with efficient vectorized evaluation once the model characteristic function is available.

Let $x = X(t_0)$ and $y = X(T)$, and consider a truncation interval $[a, b]$ that concentrates most of the probability mass of $X(T) \mid X(t_0) = x$. On this interval, the conditional density is approximated by

$$\hat{f}_X(y) = \sum_{k=0}^{N-1}{}' \bar{F}_k \cos\left(\pi k \frac{y-a}{b-a}\right), \quad (2.19)$$

where $\hat{f}_X$ is the truncated cosine approximation of the density, N is the number of cosine terms, k indexes the expansion terms, $\sum'$ indicates that the $k = 0$ term is weighted by $1/2$, and the cosine coefficients are obtained from the characteristic function:

$$\bar{F}_k := \frac{2}{b-a} \Re \left[\phi_X \left(\frac{\pi k}{b-a} \right) \exp \left(-\frac{i\pi a k}{b-a} \right) \right]. \quad (2.20)$$

The operator $\Re[\cdot]$ denotes the real part.

Starting from

$$V(t_0, x) = e^{-r\tau} \int_{\mathbb{R}} V(T, y) \hat{f}_X(T, y; t_0, x) dy, \quad (2.21)$$

the COS discretization yields

$$V(t_0, x) \approx e^{-r\tau} \sum_{k=0}^{N-1}{}' \Re \left[\phi_X \left(\frac{\pi k}{b-a} \right) \exp \left(-\frac{i\pi a k}{b-a} \right) \right] H_k, \quad (2.22)$$

where the payoff coefficients are

$$H_k := \frac{2}{b-a} \int_a^b V(T, y) \cos\left(\frac{y-a}{b-a} \pi k\right) dy. \quad (2.23)$$

For plain-vanilla options, H_k has closed-form expressions, so the numerical effort is concentrated in evaluating the characteristic function on a fixed frequency grid [2, Sec. 3].

The choice of $[a, b]$ controls truncation error. Following the cumulant-based rule in [2, Sec. 5], we use

$$[a, b] = \left[c_1 - L \sqrt{c_2 + \sqrt{c_4}}, c_1 + L \sqrt{c_2 + \sqrt{c_4}} \right], \quad (2.24)$$

with c_1, c_2, c_4 the first, second, and fourth cumulants of $X(T)$ and $L > 0$ a scaling parameter. This connects directly with the cumulant machinery introduced in Section 2.3. In the numerical experiments of this thesis, the integration interval is selected using the refined cumulant-based strategy detailed in Appendix A.

From a numerical viewpoint, it is useful to separate two error sources:

$$|V - V_N^{\text{COS}}| \leq \underbrace{\varepsilon_{\text{trunc}}([a, b])}_{\text{domain truncation}} + \underbrace{\varepsilon_N}_{\text{finite cosine series}}. \quad (2.25)$$

Here, V_N^{COS} is the COS approximation using N terms, $\varepsilon_{\text{trunc}}$ is the error from truncating the integration domain to $[a, b]$, and ε_N is the finite-series error. Increasing L reduces $\varepsilon_{\text{trunc}}$ but may require larger N to keep ε_N small; reducing L has the opposite effect. In practice, we choose (L, N) jointly and verify stability, especially for extreme strikes and short maturities, where numerical artifacts appear first.

Implementation is fully vectorized: frequencies $u_k = \pi k / (b - a)$, exponential phase factors, and payoff coefficients are precomputed once per maturity, then reused across strike grids. This is one of the reasons COS is well suited for repeated-pricing workflows such as calibration loops and synthetic dataset generation.

2.6.1. The COS Method Applied To The Heston Model

Following the notation of Sections 2.5–2.6 and [2, Sec. 3], define

$$u_k := \frac{k\pi}{b-a}, \quad x := \log\left(\frac{S_0}{K}\right), \quad y := \log\left(\frac{S_T}{K}\right).$$

Here, x and y are log-moneyness variables: they measure the underlying price relative to the strike before and at maturity. Positive log-moneyness means the underlying is above the strike, while negative log-moneyness means it is below the strike. Under Heston dynamics, the conditional characteristic function of y can be written in affine form [1], [13]

$$\varphi_H(u; \tau, x, v_0) = \exp(C(u, \tau) + D(u, \tau)v_0 + iux), \quad (2.26)$$

with

$$d(u) = \sqrt{(\kappa - i\rho\gamma u)^2 + \gamma^2(u^2 + iu)}, \quad (2.27)$$

$$g(u) = \frac{\kappa - i\rho\gamma u - d(u)}{\kappa - i\rho\gamma u + d(u)}, \quad (2.28)$$

$$C(u, \tau) = iur\tau + \frac{\kappa\bar{v}}{\gamma^2} \left[(\kappa - i\rho\gamma u - d(u))\tau - 2 \log\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right) \right], \quad (2.29)$$

$$D(u, \tau) = \frac{\kappa - i\rho\gamma u - d(u)}{\gamma^2} \left(\frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}} \right). \quad (2.30)$$

Here, φ_H denotes the Heston characteristic function of the log-moneyness state, while C , D , d , and g are complex-valued auxiliary functions of the Fourier variable u and maturity τ . This is the standard semi-closed representation used in Fourier pricing; numerically, consistency in branch selection for $d(u)$ and the complex logarithm is required for stability [13, Chs. 2–3].

The COS price for one strike can then be written as

$$V(t_0, x) \approx K e^{-r\tau} \sum_{k=0}^{N-1} \Re \left[\varphi_H(u_k; \tau, x, \nu_0) e^{-iu_k a} \right] U_k, \quad (2.31)$$

where U_k are payoff-dependent COS coefficients. For plain-vanilla options:

$$U_k^{\text{call}} = \frac{2}{b-a} (\chi_k(0, b) - \psi_k(0, b)), \quad U_k^{\text{put}} = \frac{2}{b-a} (-\chi_k(a, 0) + \psi_k(a, 0)), \quad (2.32)$$

with

$$\chi_k(c, d) = \frac{1}{1 + u_k^2} \left[e^d (\cos(u_k(d-a)) + u_k \sin(u_k(d-a))) - e^c (\cos(u_k(c-a)) + u_k \sin(u_k(c-a))) \right], \quad (2.33)$$

$$\psi_k(c, d) = \begin{cases} d - c, & k = 0, \\ \frac{b-a}{k\pi} [\sin(u_k(d-a)) - \sin(u_k(c-a))], & k \geq 1. \end{cases} \quad (2.34)$$

Here, χ_k and ψ_k are analytical payoff integrals over generic bounds c and d inside the truncation interval. In the implementation of this thesis, data generation is run on European puts as a numerical convention, mainly associated with the COS-pricing setup used in the project configuration.

This subsection establishes the analytical Heston-COS pricing block used as the reference framework in the following chapters.

2.6.2. Semi-Analytical Greeks from the Heston Characteristic Function

For validation purposes, Greeks can also be defined in the semi-analytical Heston framework, i.e., directly from the characteristic-function pricing representation. For a European call, a standard form is [1] and is developed in [13, Chapter 1]

$$C(S_0, K, \tau) = S_0 \Pi_1 - K e^{-r\tau} \Pi_2, \quad (2.35)$$

where $C(S_0, K, \tau)$ is the European call price and Π_1, Π_2 are risk-neutral probability-type terms in the Heston pricing formula:

$$\Pi_j = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \Re \left(\frac{e^{-iu \log K} \Psi_j(u)}{iu} \right) du, \quad j \in \{1, 2\}, \quad (2.36)$$

with

$$\Psi_2(u) = \phi_X(u), \quad \Psi_1(u) = \frac{\phi_X(u-i)}{\phi_X(-i)}, \quad (2.37)$$

where j selects the two probability terms, Ψ_j are the corresponding Fourier integrands, and ϕ_X is the characteristic function of $X_T = \log S_T$ under Heston [1]; see also [13, Chapter 1].

In this context, Greeks are sensitivities of the option value to market or model inputs. Delta measures sensitivity to the underlying price, Gamma measures the curvature of that sensitivity, Vega measures sensitivity to volatility or variance, Rho measures sensitivity to the interest rate, and Theta measures sensitivity to the passage of time.

Let p denote a generic model/input variable ($r, \tau, v_0, \rho, \kappa, \gamma, \bar{v}$, etc.). Under standard regularity and integrability conditions for affine models, differentiation under the integral sign yields [13, Chapter 11]

$$\partial_p \Pi_j = \frac{1}{\pi} \int_0^\infty \Re \left(\frac{e^{-iu \log K}}{iu} \partial_p \Psi_j(u) \right) du. \quad (2.38)$$

Therefore, Greeks are obtained by combining the derivatives of Π_1, Π_2 through the pricing identity [13, Chapter 11]. For example:

$$\Delta_{\text{call}} = \partial_{S_0} C = \Pi_1 + S_0 \partial_{S_0} \Pi_1 - K e^{-r\tau} \partial_{S_0} \Pi_2, \quad (2.39)$$

$$\Gamma_{\text{call}} = \partial_{S_0}^2 C = 2 \partial_{S_0} \Pi_1 + S_0 \partial_{S_0}^2 \Pi_1 - K e^{-r\tau} \partial_{S_0}^2 \Pi_2, \quad (2.40)$$

$$\text{Vega}_{v_0} = \partial_{v_0} C = S_0 \partial_{v_0} \Pi_1 - K e^{-r\tau} \partial_{v_0} \Pi_2, \quad (2.41)$$

$$\text{Rho} = \partial_r C = S_0 \partial_r \Pi_1 + K \tau e^{-r\tau} \Pi_2 - K e^{-r\tau} \partial_r \Pi_2, \quad (2.42)$$

$$\Theta = \partial_t C = -\partial_\tau C. \quad (2.43)$$

The practical advantage is that all sensitivities are expressed in terms of derivatives of the same characteristic-function integrands. In affine form, $\phi_X(u) = \exp(C(u, \tau) + D(u, \tau)v_0 + iu \log S_0)$, so parameter derivatives follow analytically from derivatives of C and D , and can be inserted directly into the integrals above.

2.7. Implied Volatility And Numerical Inversion

For each target option value V_{target} (market quote or model-generated price), implied volatility is obtained by inverting the Black–Scholes map [5]. This provides a common representation across strikes and maturities, which is the main comparison space used in this thesis.

Numerically, we solve the scalar nonlinear equation

$$F(\sigma) := V_{BS}(S_0, K, r, \tau, \sigma) - V_{\text{target}} = 0, \quad \sigma > 0. \quad (2.44)$$

For European vanilla options, V_{BS} is strictly increasing in σ when $\tau > 0$, so the root is unique whenever V_{target} satisfies static no-arbitrage bounds.

The inversion routine used in this thesis is the bracketed Brent method [14, Ch. 9]. This choice is robust for the one-dimensional IV root and avoids derivative-sensitivity issues in low-vega regions. As an alternative, Levenberg–Marquardt was also tested in preliminary experiments, but it showed lower numerical stability in this workflow; therefore, it is not used in the final pipeline.

Stopping is based on standard price and parameter tolerances:

$$|F(\sigma_n)| \leq \varepsilon_{\text{price}} \quad \text{or} \quad |\sigma_{n+1} - \sigma_n| \leq \varepsilon_{\sigma}, \quad n \leq n_{\text{max}}. \quad (2.45)$$

Here, σ_n is the volatility iterate at step n , $\varepsilon_{\text{price}}$ and ε_{σ} are stopping tolerances, and n_{max} is the maximum number of iterations. If convergence is not reached, or if the target price violates arbitrage bounds, the point is flagged as invalid for IV reporting. In edge cases near the IV floor, the pricing stage may be recomputed with a wider COS truncation interval before retrying inversion.

In the results chapters, IV errors are reported with a unified metric set:

$$\begin{aligned} \text{MSE}_{IV} &= \frac{1}{N} \sum_{i=1}^N (\hat{\sigma}_i - \sigma_i^{\text{ref}})^2, \\ \text{RMSE}_{IV} &= \sqrt{\text{MSE}_{IV}}, \\ \text{MAPE}_{IV} &= 100 \frac{1}{N} \sum_{i=1}^N \frac{|\hat{\sigma}_i - \sigma_i^{\text{ref}}|}{\max(|\sigma_i^{\text{ref}}|, \varepsilon_{\text{MAPE}})}. \end{aligned} \quad (2.46)$$

Here, MSE_{IV} and RMSE_{IV} measure squared and root-squared error in IV units, while MAPE_{IV} gives a scale-relative percentage error with denominator floor $\varepsilon_{\text{MAPE}}$ to avoid unstable ratios near zero. The same convention is reused for price and Greek benchmarks by replacing σ with the corresponding target variable. Tail quantiles may be used as auxiliary diagnostics, but not as primary benchmark metrics.

3. NEURAL NETWORK PRICER FOR HESTON

The main objective of this chapter is to construct an ANN-based surrogate pricer for European vanilla options under the Heston stochastic-volatility model. The target quantity is implied volatility, and the surrogate is designed to approximate the forward map

$$(\Theta, m, \tau, r) \mapsto \sigma_{imp}.$$

Reference pricing based on numerical methods (in our pipeline, Heston COS valuation plus implied-volatility inversion) is highly accurate but computationally expensive when repeated many times, as required in calibration and sensitivity workflows. The ANN aims to preserve pricing accuracy while reducing inference time by several orders of magnitude.

Methodologically, this chapter follows the ANN-pricing framework of Liu et al. [3] as the main reference (architecture family, data ranges, and training philosophy), while explicitly documenting the implementation choices adopted in this project.

This chapter focuses exclusively on the pricing network; the calibration architecture is treated separately in Chapter 4.

3.1. Problem Formulation

In this work, vanilla option pricing under Heston is posed as a supervised regression problem. Given a dataset

$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N, \quad y_i = \sigma_{imp,i},$$

the neural network approximates the map

$$f_{NN} : \mathbb{R}^8 \rightarrow \mathbb{R}, \quad \mathbf{x} = (\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r) \mapsto \sigma_{imp},$$

where $m = S_0/K$ is moneyness and $\tau = T - t$ is time-to-maturity. Each label σ_{imp} is obtained from Heston-consistent synthetic prices through numerical inversion.

The training objective is to estimate parameters $\mathbf{w}$ such that the empirical prediction error over $\mathcal{D}$ is minimized. In this chapter, we use the ANN only as a pricing surrogate, i.e., as a fast approximation of the reference numerical pipeline in the forward map $(\Theta, m, \tau, r) \mapsto \sigma_{imp}$.

We learn implied volatility instead of option price because the IV representation is directly comparable across contracts with different maturities and moneyness levels, and it is the market-standard quotation format for calibration and diagnostics. In contrast, raw prices mix scale effects (spot/strike/time) that make learning less homogeneous over the domain.

The resulting surrogate is primarily an in-domain approximator. Preliminary out-of-domain checks indicate that the network remains stable for moderate departures from the training bounds, while errors increase in extreme regimes. For this reason, the methodological core of this chapter is developed on the predefined training domain, and the extrapolation behaviour is discussed separately in the generalization and robustness analysis.

3.2. Synthetic Data Formulation

Training requires a large set of labeled input-output pairs. To obtain a controlled and fully reproducible dataset, we use synthetic data generated under the Heston model. For each sampled input vector, the target implied volatility is computed through the numerical pricing pipeline (COS valuation followed by implied-volatility inversion).

In the current pipeline, data generation is organized as a *master dataset* shared across downstream models. Each row stores:

- features: $(\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r)$,
- targets: $(price_{\text{cos}}, \sigma_{\text{imp}}^{\text{brent}})$,
- quality labels: $(feller_slack, feller_ok, brent_success, brent_abs_residual)$.

This design allows ANN and PINN workflows to reuse the same synthetic source while applying experiment-specific filters only at training time.

The raw variable set includes Heston parameters $\Theta = \{\rho, \kappa, \gamma, \bar{v}, v_0\}$ and contract/market variables $\Sigma = \{K, S_0, \tau, r\}$, which defines a 9-dimensional space. In our setup, we fix $K = 1$ and represent the underlying level through moneyness $m = S_0/K$, leading to an 8-dimensional input vector:

$$\mathbf{x} = (\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r) \in \mathbb{R}^8.$$

A full Cartesian grid is computationally prohibitive in this dimension, since using N points per variable scales as N^8 (already intractable for moderate N). Therefore, the synthetic inputs are sampled with Latin Hypercube Sampling (LHS), which stratifies each coordinate and provides broad marginal coverage with a fixed budget [15]. In our implementation, LHS is applied directly to the full input vector with a fixed seed and a target size of 10^7 samples in the master dataset. The specific ranges, constraints, and splitting/preprocessing rules are detailed below.

3.2.1. Parameter Ranges And Constraints

The sampling domain combines Heston parameters, contract variables, and market rate. Following the ranges used in our configuration (largely aligned with [3]), the training

domain is summarized in Table 3.1.

TABLE 3.1. INPUT RANGES USED FOR SYNTHETIC DATA GENERATION

Parameter	Range
ρ	$[-0.9, 0.0]$
κ	$[0.0, 3.0]$
γ	$[0.01, 0.8]$
$\bar{v}$	$[0.01, 0.5]$
v_0	$[0.05, 0.5]$
m	$[0.6, 1.4]$
τ	$[0.05, 3.0]$
r	$[0.0, 0.05]$

Source: Based on [3] and project configuration choices

To avoid non-physical or numerically unstable samples, we enforce $\tau > 0$ and monitor a Feller-type admissibility condition

$$2\kappa\bar{v} - \gamma^2 \geq 0,$$

through the scalar label

$$feller_slack = 2\kappa\bar{v} - \gamma^2,$$

and the binary indicator $feller_ok = \mathbb{1}\{feller_slack \geq 0\}$. Importantly, Feller-violating points are *not* removed during generation; they are kept in the master dataset for controlled ablation studies.

During synthesis we fix the strike to $K = 1$ and price European put options. Fixing K is equivalent to expressing spot through moneyness, which reduces the dimensionality from nine variables $(\Theta, S_0, K, \tau, r)$ to eight inputs (Θ, m, τ, r) without loss of information for this setup.

3.2.2. Dataset Splitting And Preprocessing

After generation, we keep all selected rows in the master dataset and split into train/validation/test subsets with proportions 0.8/0.1/0.1. In the reported configuration, splits are random (without mandatory stratification), and filtering policies are deferred to each training experiment.

In the baseline configuration, no explicit feature or target normalization is applied (Liu-style setup). Therefore, the model is trained directly on the physical scale of the variables $(\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r)$ and predicts IV in absolute units.

Data quality control is integrated in the synthesis pipeline through explicit labels, not hard row deletion by default. In particular, we track Brent inversion status and reconstruction error

$$|V_{BS}(\sigma_{imp}) - V_{target}|,$$

while keeping all generated points available for later filtering. For storage efficiency, output tables are persisted in `float32`, whereas COS pricing and Brent inversion are computed internally in `float64`.

3.2.3. Numerical Label Generation (COS + Brent)

Each label is produced in two deterministic numerical steps. First, the target option value V_{target} is computed with the Heston COS solver for a given input vector $(\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r)$. Second, implied volatility is obtained by solving

$$V_{BS}(\sigma) = V_{target}$$

with a Brent root-finding routine under fixed bounds and tolerance [14].

In our implementation, Brent is configured with strict numerical controls (absolute tolerance 10^{-8} , bounded search interval, and iteration cap). Additionally, when the inferred volatility hits the lower IV floor, an adaptive COS fallback can be triggered: the truncation width is increased ($L \leftarrow 1.5L$, with optional N scaling) and inversion is retried. The retry is accepted only if it improves numerical quality (e.g., leaves the floor or reduces reconstruction error).

Rather than dropping those edge cases at generation time, the pipeline records quality metadata (`brent_success`, `brent_abs_residual`, and retry diagnostics) so that robustness filters can be applied transparently at training time. This preserves full traceability of numerical reliability across the domain.

3.3. Neural Network Architecture

The pricing surrogate is implemented as a fully connected neural network that approximates a smooth nonlinear map from model/contract inputs to implied volatility. This choice is consistent with the structure of the problem: inputs are low-dimensional continuous features without spatial topology or sequential dependence, and the target is a scalar regression value.

From a function-approximation perspective, a feed-forward multilayer perceptron (MLP) provides enough expressive capacity to capture the interactions between Heston parameters, moneyness, maturity, and rates on the selected domain. Compared with architectures designed for images or sequences (e.g., CNNs or transformers), an MLP is more parsimonious and easier to train for this tabular setting.

This choice is also consistent with the classical universal-approximation results: under standard assumptions, feed-forward networks with non-polynomial activations can approximate continuous functions on compact sets to arbitrary accuracy [16], [17]. In this thesis, this theorem is used as a representation-capacity motivation, while practical accuracy is established empirically through out-of-sample benchmarks.

3.3.1. Model Class

Following [3], we use a fully connected feed-forward network (FCNN). Let $\mathbf{x} \in \mathbb{R}^8$ be

$$\mathbf{x} = (\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r),$$

and let $y = \sigma_{imp} \in \mathbb{R}$ denote the target implied volatility. The model defines a parametric function

$$\hat{\sigma}_{imp} = f_{NN}(\mathbf{x}; \mathbf{w}),$$

with trainable parameters $\mathbf{w}$.

Once $\mathbf{w}$ is fixed after training, inference is deterministic: for the same input $\mathbf{x}$, the network always returns the same prediction $\hat{\sigma}_{imp}$. This property is important for downstream calibration and sensitivity analysis, where numerical consistency across repeated calls is required.

The FCNN acts as a surrogate only at inference time. Target labels are still generated with the numerical pipeline (Heston COS pricing followed by implied-volatility inversion), so the network learns to emulate that reference solver on the sampled domain.

3.3.2. Architecture Details

The selected architecture used in the final pipeline is

$$8 \rightarrow 200 \rightarrow 200 \rightarrow 200 \rightarrow 200 \rightarrow 1,$$

with $\tanh$ activation in hidden layers and linear activation in the output layer. In compact form,

$$h^{(1)} = \phi(W^{(1)}x + b^{(1)}), \quad \dots, \quad h^{(4)} = \phi(W^{(4)}h^{(3)} + b^{(4)}), \quad \hat{y} = W^{(5)}h^{(4)} + b^{(5)},$$

where $\phi(\cdot) = \tanh(\cdot)$.

Weights are initialized with Xavier uniform initialization and biases are initialized to zero. In the baseline setup, dropout is disabled ($p = 0$) and no batch normalization layers are used. This configuration follows a controlled, reproducible design intended to isolate the impact of optimization and data quality choices.

TABLE 3.2. NEURAL NETWORK ARCHITECTURE USED FOR THE ANN PRICER

Hyperparameter	Value
Input dimension	8
Hidden layers	4
Hidden width	200 neurons per layer
Hidden activation	$\tanh$
Output dimension	1
Output activation	Linear
Weight initialization	Xavier uniform
Bias initialization	Zeros
Dropout	$p = 0.0$
Batch normalization	Disabled

Source: Based on [3], with selected $\tanh$ activation

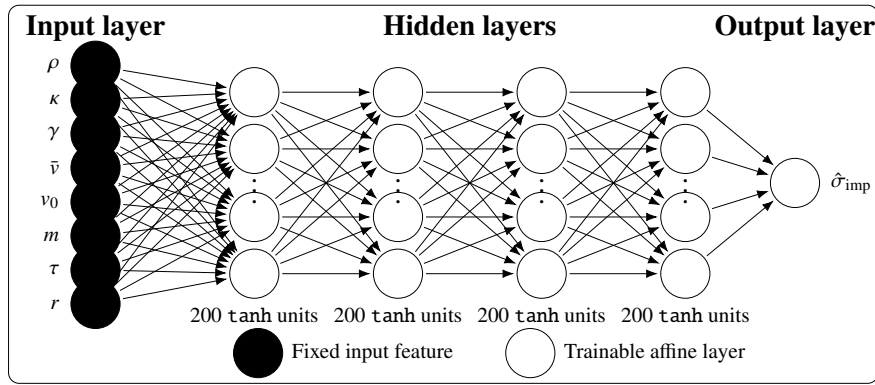


Fig. 3.1. Fully connected ANN pricer used in the forward Heston implied-volatility surrogate. The eight fixed inputs are mapped through four $\tanh$ hidden layers of width 200, and the final linear layer returns the predicted implied volatility.

3.3.3. Output Interpretation

The network output is a scalar implied volatility in annualized units (decimal form), not an option price. Therefore, prediction errors are interpreted directly in IV space, which is the representation used in the calibration workflow and in most volatility-surface diagnostics.

Because the output layer is linear, the model is not hard-constrained to produce strictly positive values for every possible input. In practice, training on a physically admissible domain with positive IV labels strongly reduces this risk inside the sampled region. For downstream use, any out-of-domain prediction can be explicitly checked and, if required, filtered before subsequent numerical steps.

This output definition also preserves differentiability of the surrogate with respect to inputs, which is useful in later chapters when the ANN pricer is embedded in calibration

and sensitivity pipelines.

3.4. Training Procedure

Training is fully configuration-driven and follows an epoch-based supervised learning loop. At each epoch, the active optimizer performs parameter updates on the training split, then the model is evaluated on the validation split without gradient computation. For Adam updates, optimization is done with standard mini-batch backpropagation; for L-BFGS updates, each step is computed through a closure that evaluates the loss and its gradient on the selected batch regime.

Each epoch stores train/validation metrics, current optimizer identity, and learning rate. When enabled, the validation monitor can be either the raw validation loss or a trimmed variant that discards the top-tail hardest samples before aggregation. This provides a robust alternative when a small fraction of outlier points may dominate the monitor value.

Execution device is selected automatically (mps when available, otherwise cpu), and data-shuffling reproducibility is controlled through a fixed generator seed from configuration. During training, two checkpoints are maintained: the *last* checkpoint (always overwritten each epoch) and the *best* checkpoint (updated only when the monitored validation metric improves). The best checkpoint is used as the reference model for post-training evaluation and downstream calibration.

3.4.1. Loss Function

Let $\{(\mathbf{x}_i, \sigma_i)\}_{i=1}^B$ denote a mini-batch, where σ_i is the target implied volatility and $\hat{\sigma}_i = f_{NN}(\mathbf{x}_i; \mathbf{w})$. The baseline objective is mean squared error (MSE):

$$\mathcal{L}_{\text{MSE}}(\mathbf{w}) = \frac{1}{B} \sum_{i=1}^B (\hat{\sigma}_i - \sigma_i)^2.$$

MSE is preferred because it is smooth, differentiable everywhere, and strongly penalizes large deviations, which is desirable when the ANN is later used in calibration loops sensitive to local pricing errors. RMSE and MAPE are then used as reporting metrics, following the convention fixed in Chapter 6.

3.4.2. Optimizer And Hyperparameters

Several optimizer and activation configurations were tested (Adam, mixed schedules, and `mix_half`). For model selection, we prioritize out-of-sample MSE/RMSE and MAPE, since this keeps the training objective and the reporting convention aligned with the benchmark chapter.

Under this criterion, the selected reference ANN is `Liu_like_tanh_1M_v01` (Adam). It preserves the Liu-style width and depth, but uses `tanh` hidden activations. This run gives the lowest validation and test MSE among the retained candidates, improves the test RMSE relative to the earlier ReLU reference, and is the ANN used in the final CaNN pipeline.

TABLE 3.3. REFERENCE AND RUNNER-UP RUNS USED IN OPTIMIZER SELECTION

Run	Activation	Optimizer	Val MSE	Test MSE	Test RMSE	Test MAPE
Liu tanh 1M v01	tanh	Adam	4.00×10^{-6}	3.47×10^{-6}	1.86×10^{-3}	15.88%
Liu ReLU v01	ReLU	Adam	4.65×10^{-6}	9.18×10^{-6}	3.03×10^{-3}	18.33%
MIX half v01	ReLU	Adam $\rightarrow$ L-BFGS	4.55×10^{-6}	9.41×10^{-6}	3.07×10^{-3}	27.17%

Source: Own elaboration from `outputs/runs/*/metrics/eval_summary.yaml`

The hyperparameters of the selected reference run (`Liu_like_tanh_1M_v01`) are summarized in Table 3.4.

TABLE 3.4. TRAINING HYPERPARAMETERS OF THE SELECTED RUN (`LIU_LIKE_TANH_1M_V01`)

Hyperparameter	Value
Training mode	Adam
Hidden activation	tanh
Seed (data-loader generator)	42
Epochs	8000
Train batch size	1024
Validation batch size	All validation samples
Loss	MSE
Adam learning rate	10^{-3}
Adam weight decay	0.0
LR scheduler	StepLR, <code>step_size = 500</code> , $\gamma = 0.5$
Early stopping	Disabled

Source: Own elaboration from run configuration copies

3.4.3. Regularization And Callbacks

Regularization in the selected run is intentionally light: dropout is disabled ($p = 0$), batch normalization is disabled, and Adam weight decay is set to zero. In addition, feature/target normalization is turned off in the Liu-style reference setup. This design keeps the training protocol close to the reference architecture and isolates the effect of the selected hidden activation, optimizer scheduling, and data quality controls.

Callback management includes: (i) checkpointing of both *best* and *last* models, (ii) StepLR scheduling for Adam, and (iii) an early-stopping module available in the pipeline but disabled in the baseline experiments. When enabled, early stopping supports both standard validation loss and trimmed validation loss monitors.

From an experimental standpoint, this callback configuration improves training robustness without over-constraining the optimization path: checkpoints protect against late-epoch degradation, while the Adam-to-L-BFGS transition plus scheduler supports stable convergence. Generalization is then assessed empirically on held-out validation and test splits rather than enforced through strong explicit regularizers.

4. NEURAL CALIBRATION AS AN INVERSE PROBLEM

The previous chapter used a neural network as a forward Heston pricer. This chapter uses the same idea in the opposite direction: given a finite implied-volatility surface, infer the Heston parameter vector

$$\theta = (\rho, \kappa, \gamma, \bar{v}, v_0).$$

Here ρ is correlation, κ is mean reversion, γ is volatility of variance, $\bar{v}$ is long-run variance, and v_0 is the initial variance. The chapter follows the neural calibration idea of Liu et al. [3]: a trained ANN pricer is frozen and used as a fast evaluator inside a calibration loop.

The purpose is not to introduce a new calibration algorithm. The contribution in this thesis is the integration of this calibration step into the ANN–PINN workflow and the diagnostic reading of its output. For that reason, this chapter keeps only the setup needed to interpret the benchmark in Chapter 6.

4.1. Calibration Setup

The ANN surrogate from Chapter 3 approximates the forward map

$$f_{NN} : (\rho, \kappa, \gamma, \bar{v}, v_0, m, \tau, r) \mapsto \sigma_{IV},$$

where $m = S_0/K$, $\tau = T - t$, r is the risk-free rate, and σ_{IV} is the implied volatility. In calibration, the quote descriptors are fixed and only θ is varied. For each candidate θ , the network is evaluated on the quote set

$$\mathcal{Q} = \{(m_j, \tau_j, r_j, \sigma_j^{\text{mkt}}, w_j)\}_{j=1}^{N_q},$$

where $w_j \geq 0$ is an optional quote weight. The selected benchmark uses $N_q = 35$ quotes on a complete 5×7 moneyness–maturity grid with constant $r = 0.03$; the exact nodes are reported in Section 6.3.

For quote j , the predicted implied volatility and residual are

$$\hat{\sigma}_j(\theta) = f_{NN}(\rho, \kappa, \gamma, \bar{v}, v_0, m_j, \tau_j, r_j), \quad e_j(\theta) = \hat{\sigma}_j(\theta) - \sigma_j^{\text{mkt}}.$$

Calibration is therefore performed in implied-volatility space, matching the ANN training target and the market convention used for volatility smiles.

The objective minimized in the calibration loop is

$$J(\theta) = \frac{1}{\sum_{j=1}^{N_q} w_j} \sum_{j=1}^{N_q} w_j e_j(\theta)^2 + \lambda R(\theta), \quad (4.1)$$

where $R(\theta)$ is an optional regularization penalty. The calibrated vector is

$$\theta^* = \arg \min_{\theta \in \mathcal{B}} J(\theta), \quad (4.2)$$

with $\mathcal{B}$ the admissible Heston parameter box. In this thesis, the bounds are aligned with the ANN training domain: $\rho \in [-0.9, 0]$, $\kappa \in [0, 3]$, $\gamma \in [0.01, 0.8]$, $\bar{v} \in [0.01, 0.5]$, and $v_0 \in [0.05, 0.5]$. This restriction keeps calibration inside the intended interpolation regime of the forward surrogate.

4.2. Search And Stability Checks

The inverse residual surface is not assumed to be convex, so the reference calibration uses Differential Evolution followed by local polishing. This is a pragmatic choice: the method is robust to initialization, does not require gradients of the neural objective, and can evaluate a population of candidates through batched ANN inference.

Regularization is used only as a stability check. The benchmark compares the no-regularization objective with a small L^2 penalty; the fit difference is negligible, so the no-regularization run is retained as the reference. The detailed numbers are reported in Section 6.3.

The selected solution is assessed at two levels. First, quote-level residuals check whether the implied-volatility surface is reproduced. Second, parameter diagnostics check whether the selected optimum is locally stable. Around a low-residual optimum, the residual data term can be read through the Gauss–Newton approximation

$$H(\theta^*) \approx 2J_e(\theta^*)^\top W J_e(\theta^*),$$

where $J_e(\theta^*)$ is the Jacobian of quote residuals with respect to θ , and W is the diagonal matrix of quote weights. Large curvature indicates directions where the quote surface reacts strongly to parameter changes; small curvature indicates flatter, less identified directions.

This reading is deliberately cautious. A small residual confirms that the calibrated vector matches the quote grid, but it does not prove global uniqueness of the Heston inverse problem. The benchmark in Chapter 6 therefore reports both fit quality and parameter-level diagnostics.

4.3. Interface With The PINN Stage

The calibrated vector becomes the market state used by the downstream PINN analysis. The structural parameters $(\rho^*, \kappa^*, \gamma^*, \bar{v}^*)$ define the Heston operator, while v_0^* is inserted as the variance-state coordinate when evaluating the trained pricing model. This connects the ANN pricer, the calibration loop, and the PINN-based pricing and Greek computation into a single pipeline.

5. PHYSICS-INFORMED NEURAL PRICING UNDER THE HESTON MODEL

This chapter reformulates vanilla option pricing under the Heston model as a PDE-constrained learning problem. In contrast to the supervised ANN surrogate of Chapter 3, the objective is no longer to emulate precomputed numerical labels, but to approximate directly the option-price function through an unsupervised Physics-Informed Neural Network (PINN). The methodology follows the general PINN philosophy and is inspired by the Heston-specific formulation proposed in [7], while adapting the state representation and notation to the conventions used in the present work.

5.1. Problem formulation

In this chapter, European option pricing under Heston is posed as the approximation of the solution of the pricing PDE rather than as a supervised regression problem. The neural network outputs the option price itself, and the training objective is to enforce that this output satisfies the Heston valuation equation together with its terminal and boundary conditions. Consequently, the core information driving the optimization is not a set of external labels, but the analytical structure of the model under the risk-neutral measure.

To avoid a notational clash between the price function and the variance state, the option value will be denoted by u , while the instantaneous variance state will be denoted by v . This distinction is important because the Heston model contains both a variance process and a pricing function, and using the same symbol for both would make the PDE formulation unnecessarily confusing. The unknown function approximated by the PINN is therefore written as

$$u_\phi(\tau, m, v, \Theta, r),$$

where ϕ denotes the trainable weights of the network, $m = S/K$ is moneyness, $\tau = T - t$ is time-to-maturity, v is the current instantaneous variance, and Θ collects the structural Heston parameters.

More precisely, the calibrated tuple obtained in the previous chapter is

$$\Theta^* = (\rho, \kappa, \gamma, \bar{v}, v_0).$$

However, the role of v_0 must be distinguished from that of the other parameters. The quantities ρ , κ , γ and $\bar{v}$ define the operator of the Heston PDE, whereas v_0 is the initial condition of the variance process. In the PDE formulation, the state variable is the generic variance level v , not the fixed initial value v_0 . Therefore, the parametric PINN is constructed to learn a family of Heston pricing solutions indexed by the structural parameters $(\rho, \kappa, \gamma, \bar{v}, r)$, while the current variance v_0 is used only when evaluating the trained

network at the present market state. The calibrated market valuation is thus recovered through

$$u_\phi(\tau, m, v_0, \rho^*, \kappa^*, \gamma^*, \bar{v}^*, r).$$

This point is conceptually important: during training, the PINN learns the solution over a domain in $(\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r)$, whereas at inference time the current market valuation is recovered by fixing the variance coordinate to the calibrated value v_0^* and the structural parameters to the calibrated tuple Θ^* .

The network is trained on collocation points sampled in the interior and on the relevant boundaries of the pricing domain. These points are not generated by simulating full paths of the Heston variance process. Instead, they are drawn directly from a bounded domain of admissible states, for instance using Latin Hypercube Sampling (LHS), and used to evaluate the PDE residual. In that sense, the method is unsupervised: the learning signal comes from the Heston structure itself, not from externally computed price labels. This is the main methodological difference with respect to the ANN surrogate developed in Chapter 3.

At a high level, the proposed workflow is the following. First, a parametric PINN is defined to approximate the price surface over a range of maturities, moneyness levels, variance states, and Heston parameters. Second, the PINN is trained by minimizing the residual of the Heston pricing PDE together with terminal and boundary mismatches. Third, once the network is trained, the price of a contract for a calibrated market configuration is obtained by evaluating the network at the corresponding $(\tau, m, v_0^*, \Theta^*, r)$. This preserves the link with the calibration pipeline while replacing label-based supervision with model-consistent unsupervised training. In this sense, the CaNN stage remains essential: it provides the market-specific parameter tuple that is consumed by the parametric PINN at inference time, without requiring a retraining of the pricing network for each new calibration.

5.1.1. PINN Theoretical Framework

A PINN can be viewed as a parametric approximation of the solution of a differential equation. In the present setting, the network u_ϕ is required to satisfy the Heston pricing PDE in the interior of the domain and to match the contractual conditions at expiry and on selected boundaries. If $\mathcal{N}$ denotes the Heston differential operator, the target problem can be summarized abstractly as

$$\mathcal{N}[u] = 0 \quad \text{in the interior domain}, \quad \mathcal{B}[u] = 0 \quad \text{on the boundaries},$$

where $\mathcal{B}$ denotes the set of terminal and boundary constraints.

The PINN replaces the unknown analytical solution by a neural approximation u_ϕ and defines a residual function

$$\mathcal{R}_\phi = \mathcal{N}[u_\phi].$$

Training then consists of minimizing a composite loss built from three sources of error: the interior PDE residual, the mismatch with the terminal payoff, and the mismatch with the selected boundary conditions. In a compact notation, the loss takes the form

$$\mathcal{L}_{\text{PINN}}(\phi) = \mathcal{L}_{\text{PDE}}(\phi) + \mathcal{L}_{\text{term}}(\phi) + \mathcal{L}_{\text{bc}}(\phi).$$

This decomposition is standard in the PINN literature and matches the structure adopted in [7].

An essential ingredient of this approach is automatic differentiation. Since the Heston PDE contains first- and second-order derivatives with respect to the state variables, the network output must remain differentiable with respect to its inputs. The required quantities are obtained by differentiating the computational graph of the network rather than by using finite-difference approximations. This is particularly convenient in the present case because the same differentiable surrogate can later be reused for the computation of Greeks.

5.2. Heston Pricing PDE

Starting from the Heston dynamics introduced in Chapter 2 (Equation 2.11), the risk-neutral version used for pricing is

$$dS_t = rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \quad (5.1)$$

$$dv_t = \kappa(\bar{v} - v_t) dt + \gamma \sqrt{v_t} dW_t^{(2)}, \quad (5.2)$$

with instantaneous correlation

$$d\langle W^{(1)}, W^{(2)} \rangle_t = \rho dt.$$

Here r is the risk-free rate, κ is the mean-reversion speed of the variance process, $\bar{v}$ is the long-run variance level, γ is the volatility of variance, and ρ controls the correlation between returns and variance shocks. The current variance level at valuation time is denoted by v_0 , so that v_t evolves from the initial condition $v(0) = v_0$.

Let $u(t, S, v)$ denote the price of a European vanilla option. By the Feynman-Kac representation, u satisfies the Heston pricing PDE [1]

$$u_t + \frac{1}{2} v S^2 u_{SS} + \rho \gamma v S u_{Sv} + \frac{1}{2} \gamma^2 v u_{vv} + r S u_S + \kappa(\bar{v} - v) u_v - r u = 0. \quad (5.3)$$

This is the continuous-time constraint that will be enforced by the PINN at interior collocation points. In the present chapter, the price function is denoted by u precisely to keep it separate from the variance state v .

For implementation purposes, it is more convenient to work with time-to-maturity $\tau = T - t$ and moneyness $m = S/K$ instead of calendar time t and spot S . Since the synthetic and market pipelines of this project use a fixed strike normalization $K = 1$, the

pricing problem can be rewritten in terms of the state variables (τ, m, v) . Defining the same pricing function in transformed coordinates, the PDE becomes

$$u_\tau = \frac{1}{2}vm^2u_{mm} + \rho\gamma vmu_{mv} + \frac{1}{2}\gamma^2vu_{vv} + rm u_m + \kappa(\bar{v} - v)u_v - ru. \quad (5.4)$$

This is the formulation that will be used to define the PDE residual in the training loss. It remains fully equivalent to the original Heston pricing equation, but it is better aligned with the representation used in the rest of the thesis.

For a European put option with normalized strike $K = 1$, the terminal condition at expiry is

$$u(0, m, v) = (1 - m)^+. \quad (5.5)$$

The payoff does not depend explicitly on the variance state, which means that the terminal condition is imposed uniformly over the v -dimension of the domain. In addition, a natural lower boundary condition is obtained at zero underlying value:

$$u(\tau, 0, v) = e^{-r\tau}. \quad (5.6)$$

For a general strike K , this condition would read $u(\tau, 0, v) = Ke^{-r\tau}$. In a first PINN specification, these two constraints are sufficient to reproduce the structure used in [7]: one term of the loss is attached to the interior PDE residual, one to the terminal payoff, and one to the lower boundary.

The far-field behavior as $m \rightarrow \infty$ for a put, namely $u(\tau, m, v) \rightarrow 0$, can also be incorporated if needed. However, to keep the initial implementation simple, the first version of the model will explicitly enforce only the terminal condition and the lower boundary, while controlling the remaining behavior through the bounded sampling domain in (τ, m, v) and the interior residual. This choice is consistent with the objective of building a minimal but coherent unsupervised PINN before adding more elaborate constraints.

5.3. Neural Network Architecture

5.3.1. Model Class

The pricing PINN is implemented as a fully connected feed-forward neural network (MLP) that approximates the mapping

$$u_\phi : \mathbb{R}^8 \rightarrow \mathbb{R}, \quad x = (\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r) \mapsto u_\phi(x),$$

where the scalar output is the option price. This architecture is appropriate for the present setting because all inputs are continuous tabular variables and no spatial or sequential inductive bias is required. This is also the standard architectural choice in many PINN implementations, where a smooth MLP acts as the differentiable trial function for the PDE solution [7], [11].

Compared with Chapter 3, where the ANN surrogate learned implied volatility from supervised labels, the role of the network here is to represent a differentiable pricing function that is constrained by the Heston PDE and boundary conditions. The model class is therefore selected not only for approximation capacity, but also for stable first- and second-order automatic differentiation. This differentiability requirement is central when PDE residuals or derivative losses are part of the objective, as in gradient-enhanced and Sobolev-style PINN variants [4], [8].

5.3.2. Architecture Details

The reference architecture used in the implementation is

$$8 \rightarrow 256 \rightarrow 256 \rightarrow 256 \rightarrow 256 \rightarrow 1,$$

with `tanh` activation in hidden layers and linear activation in the output layer. In compact notation,

$$h^{(1)} = \varphi(W^{(1)}x + b^{(1)}), \dots, h^{(4)} = \varphi(W^{(4)}h^{(3)} + b^{(4)}), \quad u_\phi = W^{(5)}h^{(4)} + b^{(5)},$$

with $\varphi = \tanh$.

Weights are initialized with Xavier uniform initialization and biases are initialized to zero. In the baseline setup, dropout is disabled and no batch-normalization layers are included.

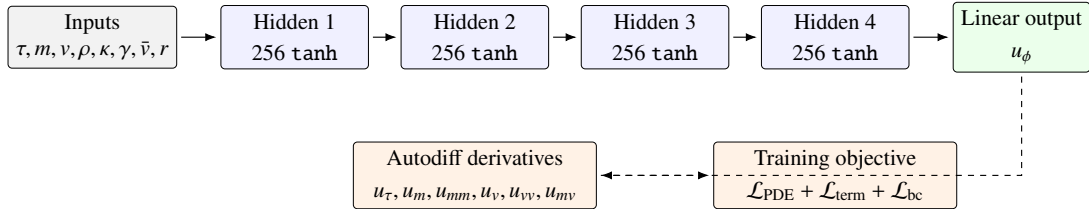


Fig. 5.1. PINN architecture used for Heston pricing. The network maps the eight state and parameter inputs to a scalar option price; automatic differentiation of the same graph provides the derivatives required by the PDE residual and by Greek computation.

TABLE 5.1. PINN ARCHITECTURE USED IN THIS CHAPTER

Hyperparameter	Value
Input dimension	8
Hidden layers	4
Hidden width	256 neurons per layer
Hidden activation	$\tanh$
Output dimension	1
Output activation	Linear
Weight initialization	Xavier uniform
Bias initialization	Zeros
Dropout	$p = 0.0$
Batch normalization	Disabled

Source: Own implementation (`configs/pinn_model_architecture.yaml`)

5.3.3. Model Inputs and Outputs

The model receives the following ordered input vector:

$$x = (\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r),$$

where (τ, m, v) are state variables, r is the risk-free rate, and $(\rho, \kappa, \gamma, \bar{v})$ are the structural Heston parameters that define the PDE operator.

The network output is a scalar price $u_\phi(x)$, corresponding to the normalized-strike vanilla contract considered in this chapter ($K = 1$). This output choice is consistent with the PDE-constrained formulation: the PINN directly approximates the pricing function rather than an implied-volatility transformation. At inference time, the CaNN calibration block supplies the market-specific values of $(\rho, \kappa, \gamma, \bar{v}, v_0)$; the PINN then evaluates the contract by fixing $v = v_0$ and varying the contract coordinates (τ, m) and the market rate r .

In the current experimental setup, collocation generation is run in a fixed-parameter regime for $(\rho, \kappa, \gamma, \bar{v})$, while (τ, m, v, r) vary across points. This means the model remains formally 8-dimensional, but some coordinates may be constant in a given run due to calibration-conditioned sampling. This is the baseline fixed- Θ benchmark used in the reported experiments; the same input convention leaves the fully parametric variant available as a direct extension.

Input Scaling

Although the pricing problem is formulated in physical variables, the optimization is performed with an affine rescaling of the network inputs. The motivation is numerical: (τ, m, v) and the Heston-related coordinates do not share comparable magnitudes, which

can deteriorate conditioning and gradient balance. Following practical PINN recommendations, centering and scaling are used as a stabilization device; this is consistent with the broader observation that conditioning and gradient-scale imbalance strongly affect PINN training dynamics [9], [10], [11].

In the implemented pipeline, the transformation is applied componentwise to the full PINN input vector through

$$\tilde{x}_d = a_d + b_d x_d, \quad d = 1, \dots, 8.$$

The coefficients are estimated from the collocation points of the training split:

$$\mu_d = \frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} x_{i,d}, \quad \sigma_d = \sqrt{\frac{1}{N_{\text{train}}} \sum_{i=1}^{N_{\text{train}}} (x_{i,d} - \mu_d)^2},$$

and then

$$a_d = -\frac{\mu_d}{\sigma_d}, \quad b_d = \frac{1}{\sigma_d}.$$

To avoid numerical issues on nearly constant coordinates, the implementation uses a small threshold $\varepsilon = 10^{-8}$: if $\sigma_d \leq \varepsilon$, that coordinate is left unscaled ($a_d = 0$, $b_d = 1$).

The PDE residual is still defined in physical variables and differentiated by automatic differentiation. In code, the network is evaluated at $\tilde{x}$, while derivatives are taken with respect to the original coordinates x ; the chain rule is therefore handled by the computational graph. Consequently, input scaling changes the numerical conditioning of training, but not the financial meaning of the PDE constraint.

5.4. Training Procedure

5.4.1. Loss Function

The PINN is trained by minimizing a composite loss that enforces the pricing PDE in the interior of the domain together with the terminal payoff and the lower boundary condition. Let $u_\phi(\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r)$ denote the output of the neural network. From the transformed Heston PDE introduced above, the residual at an interior point is defined as

$$\mathcal{R}_\phi = u_\tau - \frac{1}{2} v m^2 u_{mm} - \rho \gamma v m u_{mv} - \frac{1}{2} \gamma^2 v u_{vv} - r m u_m - \kappa (\bar{v} - v) u_v + r u, \quad (5.7)$$

where all derivatives are evaluated by automatic differentiation on the computational graph of the network. In the ideal case, the exact pricing function would satisfy $\mathcal{R}_\phi = 0$ for every admissible interior point.

Let $\mathcal{S}_D = \{z_j\}_{j=1}^{N_D}$ denote the set of interior collocation points, with

$$z_j = (\tau_j, m_j, v_j, \rho_j, \kappa_j, \gamma_j, \bar{v}_j, r_j).$$

The interior loss is the mean squared residual

$$\mathcal{L}_{\text{PDE}}(\phi) = \frac{1}{N_D} \sum_{j=1}^{N_D} \mathcal{R}_\phi(z_j)^2. \quad (5.8)$$

On the terminal set $\mathcal{S}_T = \{z_k^T\}_{k=1}^{N_T}$, where $\tau = 0$, the network must match the payoff of the European put. The corresponding error is

$$e_k^T(\phi) = u_\phi(0, m_k, v_k, \rho_k, \kappa_k, \gamma_k, \bar{v}_k, r_k) - (1 - m_k)^+, \quad (5.9)$$

and the terminal loss is defined by

$$\mathcal{L}_{\text{term}}(\phi) = \frac{1}{N_T} \sum_{k=1}^{N_T} \left(e_k^T(\phi) \right)^2. \quad (5.10)$$

On the lower boundary set $\mathcal{S}_{\text{low}} = \{z_l^{\text{low}}\}_{l=1}^{N_{\text{low}}}$, where $m = 0$, the exact put price is the discounted strike. Since the implementation uses the normalized strike $K = 1$, the lower-boundary error is

$$e_l^{\text{low}}(\phi) = u_\phi(\tau_l, 0, v_l, \rho_l, \kappa_l, \gamma_l, \bar{v}_l, r_l) - e^{-r_l \tau_l}, \quad (5.11)$$

and the associated loss is

$$\mathcal{L}_{\text{bc}}(\phi) = \frac{1}{N_{\text{low}}} \sum_{l=1}^{N_{\text{low}}} \left(e_l^{\text{low}}(\phi) \right)^2. \quad (5.12)$$

The total training objective is then written as

$$\mathcal{L}_{\text{PINN}}(\phi) = \mathcal{L}_{\text{PDE}}(\phi) + \mathcal{L}_{\text{term}}(\phi) + \mathcal{L}_{\text{bc}}(\phi). \quad (5.13)$$

For implementation, it is convenient to keep an explicit weighted form,

$$\mathcal{L}_{\text{PINN}}(\phi) = \lambda_{\text{PDE}} \mathcal{L}_{\text{PDE}}(\phi) + \lambda_{\text{term}} \mathcal{L}_{\text{term}}(\phi) + \lambda_{\text{low}} \mathcal{L}_{\text{bc}}(\phi), \quad (5.14)$$

with non-negative coefficients $(\lambda_{\text{PDE}}, \lambda_{\text{term}}, \lambda_{\text{low}})$ that can be tuned if one component dominates the optimization. In the baseline specification of this thesis, all three weights are initialized to one, i.e. $(1, 1, 1)$, following the simplest formulation in [7].

A practical point is that the PDE term requires second-order derivatives with respect to inputs. Therefore, the automatic-differentiation graph must be kept when computing first derivatives, so that mixed and second derivatives such as u_{mv} and u_{vv} can be obtained consistently in the same computational graph. This is the key implementation detail that enables direct optimization of the PDE residual without finite-difference approximations.

5.4.2. Optimizer and Hyperparameters

The optimization follows a two-stage strategy that is common in PINN practice: an initial stochastic phase with Adam, followed by a quasi-Newton refinement with L-BFGS. In this work, this strategy is implemented as a mixed schedule in which Adam is used during the first part of training and L-BFGS in the second part, while a pure Adam or pure L-BFGS run remains available as an ablation.

Since the three loss terms are evaluated on different sample sets, each epoch uses three mini-batches: one from $\mathcal{S}_D$ (interior), one from $\mathcal{S}_T$ (terminal), and one from $\mathcal{S}_{\text{low}}$ (lower boundary). This allows different batch sizes for interior and boundary constraints and keeps the training loop aligned with the composite objective.

The reference hyperparameter block for the first implementation includes: learning rates and optimizer-specific settings, the Adam-to-L-BFGS switching epoch in mixed mode, batch sizes for interior and boundary sets, and the triplet of loss weights $(\lambda_{\text{PDE}}, \lambda_{\text{term}}, \lambda_{\text{low}})$.

For diagnostics, the training run stores epoch-wise train/validation curves for the total weighted loss and for each component separately. In addition, a two-dimensional error map over (m, τ) is produced from validation interior points by averaging $|\mathcal{R}_\phi|$ in bins. This map helps identify whether specific regions of the contract domain remain systematically underfitted even when the global loss decreases.

5.4.3. Sampling strategy

Training requires three different sample sets, each associated with one component of the loss. As in the general PINN methodology, these sets are not built from labeled prices but from points drawn directly in the state-parameter domain. The first set corresponds to the interior of the PDE domain, the second to the terminal boundary at expiry, and the third to the lower boundary at zero underlying value.

The interior domain is defined over bounded intervals for the state variables and for the Heston parameters:

$$\tau \in [0, \tau_{\max}], \quad m \in [m_{\min}, m_{\max}], \quad v \in [v_{\min}, v_{\max}],$$

combined with admissible ranges for $(\rho, \kappa, \gamma, \bar{v}, r)$. These parameter intervals should remain consistent with the calibration regime studied in the previous chapters, since the purpose of the parametric PINN is precisely to generalize over the family of Heston configurations that may be produced by the CaNN stage. In particular, the v -dimension is sampled explicitly as a state coordinate of the PDE. This is a crucial point: the PINN is trained on generic variance states v , whereas the calibrated initial variance v_0 is only used later, when evaluating the trained network at the current market state.

To obtain a representative coverage of this multidimensional domain with a controlled number of points, the baseline sampling method will be Latin Hypercube Sam-

pling (LHS). Compared with naive Monte Carlo sampling, LHS provides a more regular coverage of each marginal dimension and reduces the risk of leaving large regions of the domain underexplored. In the present setting, this is particularly useful because the PINN must generalize simultaneously over state variables and model parameters.

For the interior set $\mathcal{S}_D$, each sampled point has the form

$$z_j = (\tau_j, m_j, v_j, \rho_j, \kappa_j, \gamma_j, \bar{v}_j, r_j), \quad j = 1, \dots, N_D.$$

These are the collocation points at which the PDE residual is evaluated. For the terminal set $\mathcal{S}_T$, points are sampled in the subdomain

$$\tau = 0, \quad m \in [m_{\min}, m_{\max}], \quad v \in [v_{\min}, v_{\max}],$$

again jointly with the Heston parameters. This set is used to impose the payoff condition. For the lower-boundary set $\mathcal{S}_{\text{low}}$, the sampled points satisfy

$$m = 0, \quad \tau \in [0, \tau_{\max}], \quad v \in [v_{\min}, v_{\max}],$$

with the same parameter ranges, and are used to enforce the discounted-strike boundary condition.

From an implementation viewpoint, the three sets can be regenerated dynamically during training or precomputed once and reused across epochs. A dynamic resampling strategy may improve coverage of the domain and reduce overfitting to a fixed cloud of collocation points, whereas a static design simplifies reproducibility and debugging. The initial implementation will prioritize simplicity and reproducibility; therefore, the first reference version will rely on fixed sample sets generated before the training loop. Dynamic resampling can later be evaluated as an extension if it leads to more stable convergence.

This sampling strategy preserves the fully unsupervised nature of the method. No prices from COS, FFT, or Monte Carlo are required to build the training objective. Numerical pricing methods remain useful for ex-post validation and benchmarking, but not as a source of labels for fitting the PINN itself.

5.5. Extension: Sobolev-Enhanced Training for Greek Accuracy

The baseline PINN objective in this chapter is designed to recover prices by enforcing the Heston PDE and boundary/terminal conditions. However, accurate prices do not automatically imply accurate first- and second-order sensitivities. In particular, Γ and variance/rate sensitivities may show larger local errors because they depend on curvature properties of the learned surface. To address this limitation, a natural extension is to enrich training with derivative-aware terms inspired by Sobolev training [4] and gradient-enhanced PINN formulations [8]. The formulation below is an application-specific hybrid: it follows Sobolev Training in matching target derivatives, but unlike gPINN it does not penalize gradients of the PDE residual directly.

The extension keeps the original physics objective and adds supervised derivative terms on a dedicated set of Greek-anchor points. Let

$$\mathcal{L}_{\text{base}} = \lambda_{\text{pde}} \mathcal{L}_{\text{pde}} + \lambda_{\text{term}} \mathcal{L}_{\text{term}} + \lambda_{\text{low}} \mathcal{L}_{\text{low}},$$

and let $\mathcal{S}_g$ be a set of anchor points in the same input space

$$x = (\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r).$$

For each $x \in \mathcal{S}_g$, reference sensitivities $\mathcal{G}^*(x)$ are obtained from the semi-analytical He-ston characteristic-function map (Section 2.6.2), while $\mathcal{G}_\phi(x)$ are computed from autodiff on the PINN output.

To prevent scale dominance across Greeks, we use normalized residuals and split supervision into two blocks:

$$\mathcal{L}_{g,1} = \frac{1}{|\mathcal{S}_g| |\mathcal{Q}_1|} \sum_{x \in \mathcal{S}_g} \sum_{q \in \mathcal{Q}_1} \left(\frac{q_\phi(x) - q^*(x)}{s_q + \varepsilon} \right)^2, \quad \mathcal{Q}_1 = \{\Delta, \text{Vega}_v, \Theta, \text{Rho}\}, \quad (5.15)$$

$$\mathcal{L}_{g,2} = \frac{1}{|\mathcal{S}_g|} \sum_{x \in \mathcal{S}_g} \left(\frac{\Gamma_\phi(x) - \Gamma^*(x)}{s_\Gamma + \varepsilon} \right)^2, \quad (5.16)$$

where s_q are fixed scale factors (e.g., robust quantiles of $|q^*|$ on $\mathcal{S}_g$) and $\varepsilon > 0$ avoids numerical singularities near zero. The separation of $\mathcal{L}_{g,2}$ for Γ gives independent control of curvature supervision, which is typically more unstable than first-order terms.

The prototype objective is then

$$\mathcal{L}_{\text{proto}} = \mathcal{L}_{\text{base}} + \lambda_{g,1} \mathcal{L}_{g,1} + \lambda_{g,2} \mathcal{L}_{g,2}. \quad (5.17)$$

From an optimization viewpoint, this extension introduces additional multi-term imbalance risks, already documented in the PINN literature [9], [10], [11]. Therefore, the intended training strategy is two-stage: first, train the baseline PINN to a stable price solution; second, apply a short fine-tuning phase with gradually increased $(\lambda_{g,1}, \lambda_{g,2})$, while monitoring possible regressions in price/PDE metrics.

This section defines a methodological prototype for subsequent work. It is included to fix the exact formulation to be tested, but it is not yet part of the benchmarked implementation reported in this thesis.

5.6. Greeks Computation

5.6.1. Definition of Greeks from the PINN Output

Once the PINN has been trained, the network output is a differentiable approximation of the vanilla option price:

$$u_\phi(\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r).$$

Therefore, sensitivities can be obtained directly as partial derivatives of this function, evaluated at a chosen market point. Let

$$x = (\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r).$$

The first-order and second-order derivatives with respect to the input coordinates are denoted by

$$\nabla_x u_\phi(x), \quad \nabla_x^2 u_\phi(x).$$

From these objects, standard trading Greeks are extracted as follows:

$$\Delta = \frac{\partial u_\phi}{\partial S}, \quad (5.18)$$

$$\Gamma = \frac{\partial^2 u_\phi}{\partial S^2}, \quad (5.19)$$

$$\Theta = \frac{\partial u_\phi}{\partial t} = -\frac{\partial u_\phi}{\partial \tau}, \quad (5.20)$$

$$\text{Rho} = \frac{\partial u_\phi}{\partial r}. \quad (5.21)$$

In this work, the spatial coordinate is moneyness $m = S/K$, not spot directly. Hence a chain-rule conversion is required:

$$\frac{\partial u_\phi}{\partial S} = \frac{\partial u_\phi}{\partial m} \frac{\partial m}{\partial S} = \frac{1}{K} \frac{\partial u_\phi}{\partial m}, \quad (5.22)$$

$$\frac{\partial^2 u_\phi}{\partial S^2} = \frac{1}{K^2} \frac{\partial^2 u_\phi}{\partial m^2}. \quad (5.23)$$

For the normalized-strike setup used in the thesis ($K = 1$), both coordinate systems coincide numerically.

Since the PINN state includes instantaneous variance v , a natural model-consistent volatility sensitivity is

$$\text{Vega}_v := \frac{\partial u_\phi}{\partial v}.$$

If needed, this can be transformed to a sensitivity with respect to volatility level $\sigma_v = \sqrt{v}$:

$$\frac{\partial u_\phi}{\partial \sigma_v} = 2 \sqrt{v} \frac{\partial u_\phi}{\partial v}.$$

In addition, the parametric PINN provides direct structural sensitivities

$$\frac{\partial u_\phi}{\partial \rho}, \quad \frac{\partial u_\phi}{\partial \kappa}, \quad \frac{\partial u_\phi}{\partial \gamma}, \quad \frac{\partial u_\phi}{\partial \bar{v}},$$

which are useful for risk decomposition and calibration diagnostics beyond classical desk Greeks.

5.6.2. Automatic Differentiation of the PINN

The derivative computation is based on automatic differentiation over the computational graph of the trained network. For a single input point x , we compute:

$$u_\phi(x), \quad J(x) = \nabla_x u_\phi(x), \quad H(x) = \nabla_x^2 u_\phi(x).$$

Because the model is differentiable end-to-end, these quantities are exact derivatives of the neural surrogate (up to floating-point precision), not finite-difference approximations.

Compared with bump-and-reprice finite differences, this approach has three practical advantages in the PINN context:

- it avoids step-size tuning and the associated truncation/cancellation trade-off;
- it provides first- and second-order sensitivities in a single coherent graph;
- it scales efficiently to batched inputs through vectorized Jacobian/Hessian evaluation.

From a financial perspective, the interpretation remains standard: each Greek is the local slope or curvature of the price surface produced by the PINN around the evaluation point. The quality of those sensitivities therefore depends on both the local smoothness of the trained network and the fidelity of the learned price function in that region. This closes the link with the Heston dynamics introduced in Chapter 2: the PINN price surface is trained through the PDE implied by Equation 2.11, and the Greeks are then derivatives of that model-consistent learned surface.

6. NEURAL NETWORK PERFORMANCE AND BENCHMARKING

6.1. Experimental Setup And Evaluation Protocol

This chapter compares the three model blocks developed in the thesis under a common evaluation protocol: ANN pricing surrogate, calibration network workflow, and PINN-based pricing and Greeks computation.

Unless explicitly stated, all reported metrics are computed on fixed evaluation sets with identical feature conventions, consistent preprocessing, and the same numerical references used in previous chapters.

To keep the empirical comparison coherent, the chapter uses three primary error metrics throughout: MSE, RMSE, and MAPE. MSE is aligned with the training objective used in the supervised components, RMSE restores the error to the natural output units, and MAPE provides a scale-relative percentage reading. For MAPE, denominators are floored by the same stabilized convention used in the benchmark scripts, so near-zero references do not create undefined ratios. Tail quantiles and maximum errors are used only as auxiliary diagnostics when discussing distributional behavior, not as headline metrics.

The minimal pre-benchmark validation protocol combines three checks: convergence with respect to (N, L) in the COS expansion, put-call parity residual control across strikes, and Black–Scholes consistency in near-constant-variance regimes.

These checks are used as a guardrail before comparing models, so that downstream differences can be attributed to model behavior rather than to numerical artifacts. For consistency across Sections 6.2–6.5, each block is analyzed with the same reading logic: global summary metrics, localized error diagnostics over relevant domains, and runtime indicators.

The protocol is fixed as follows. Randomness is controlled with a common seed (42) in data generation and training configurations; in calibration, the Differential Evolution seed follows the same base value unless explicitly overridden. The Heston domain is shared across experiments, with $\rho \in [-0.9, 0]$, $\kappa \in [0, 3]$, $\gamma \in [0.01, 0.8]$, $\bar{v} \in [0.01, 0.5]$, $v_0 \in [0.05, 0.5]$, and contract coordinates $m \in [0.6, 1.4]$, $\tau \in [0.05, 3.0]$ for ANN/CaNN benchmarks. Reference numerics are also fixed: COS pricing with $N = 1500$, $L = 50$, and fallback $L = 80$ when required by stability checks, plus Brent IV inversion with $\text{tol} = 10^{-8}$ and bounded search interval.

Reported metrics are always computed on fixed held-out sets, with ANN split policy fixed to 0.8/0.1/0.1 (train/validation/test), and values reported in decimal units (IV for ANN/CaNN, price for PINN). Finally, each model block reports equivalent evidence lay-

ers (global table, regional/spatial diagnostic, and convergence or runtime indicator), so cross-model comparisons remain methodologically aligned.

Methodological details of each block are kept in their dedicated chapters (Chapter 3 for ANN, Chapter 4 for CaNN, and Chapter 5 for PINN), while this chapter focuses on consolidated benchmark outcomes under the protocol above.

6.2. ANN Pricer Performance

This section reports the performance of the supervised ANN pricer introduced in Chapter 3. The objective is to assess mean implied-volatility accuracy and spatial stability over (m, τ) , using a fully specified evaluation protocol that can be reproduced without implementation-specific references.

The ANN analyzed here corresponds to the selected tanh configuration from Chapter 3. The benchmark uses fixed weights (no retraining in Chapter 6) and fixed train/-validation/test splits of sizes $8.0 \times 10^5/1.0 \times 10^5/1.0 \times 10^5$. Pointwise error is defined as

$$e = \hat{\sigma}_{IV} - \sigma_{IV}^{\text{ref}},$$

where $\hat{\sigma}_{IV}$ is the ANN-predicted implied volatility and σ_{IV}^{ref} is the reference implied volatility obtained from the Heston-COS/Brent pipeline. The error e is reported in decimal implied-volatility units.

Global quality is summarized with MSE, RMSE, and MAPE. Regional diagnostics use fixed bins over m and τ , and extrapolation behavior is analyzed through sensitivity maps with explicit interpolation limits (Figure 6.1).

6.2.1. Global Error Metrics

Table 6.1 reports split-wise global errors for the selected ANN baseline.

TABLE 6.1. GLOBAL ANN IMPLIED-VOLATILITY ERRORS
(SELECTED TANH CONFIGURATION).

Split	n	MSE	RMSE	MAPE (%)
Train	8.0×10^5	6.99×10^{-8}	2.64×10^{-4}	2.21
Validation	1.0×10^5	4.00×10^{-6}	2.00×10^{-3}	18.12
Test	1.0×10^5	3.47×10^{-6}	1.86×10^{-3}	15.88

Source: Own elaboration

The test RMSE remains in the 10^{-3} range in absolute IV units, with a lower test MSE/RMSE than the earlier ReLU reference run. MAPE is larger than the raw IV error

scale suggests because relative errors are amplified in low-IV regions; therefore it is read jointly with the spatial diagnostics below rather than as an isolated scalar.

Figure 6.1 provides a panel-wise diagnostic. Each row corresponds to one parameter plane: top $(\gamma, \bar{v})$ (vol-of-vol versus long-run variance), middle (ρ, γ) (correlation versus vol-of-vol), and bottom $(\kappa, \bar{v})$ (mean reversion versus long-run variance). The left column shows ANN IV maps with white isoclines, while the right column shows $\log |NN - \text{Heston}|$ on the same grids, using lighter tones for lower errors and darker tones for larger errors. The dashed black lines delimit interpolation limits; regions beyond those limits are extrapolation zones where the network was not trained.

The most stable behavior is observed in the interior of each plane, where isoclines are smooth and error colors remain comparatively light. Error increases near boundary segments and especially outside the dashed limits, with the strongest deterioration concentrated in outer corners of the tested planes. This is consistent with the regional-error reading: the surrogate remains accurate over the interpolation core, while low-moneyness/short-maturity-like boundary regimes are the most fragile. In practical terms, Figure 6.1 supports using the ANN as a robust interpolation surrogate and treating extrapolated regions as lower-confidence zones.

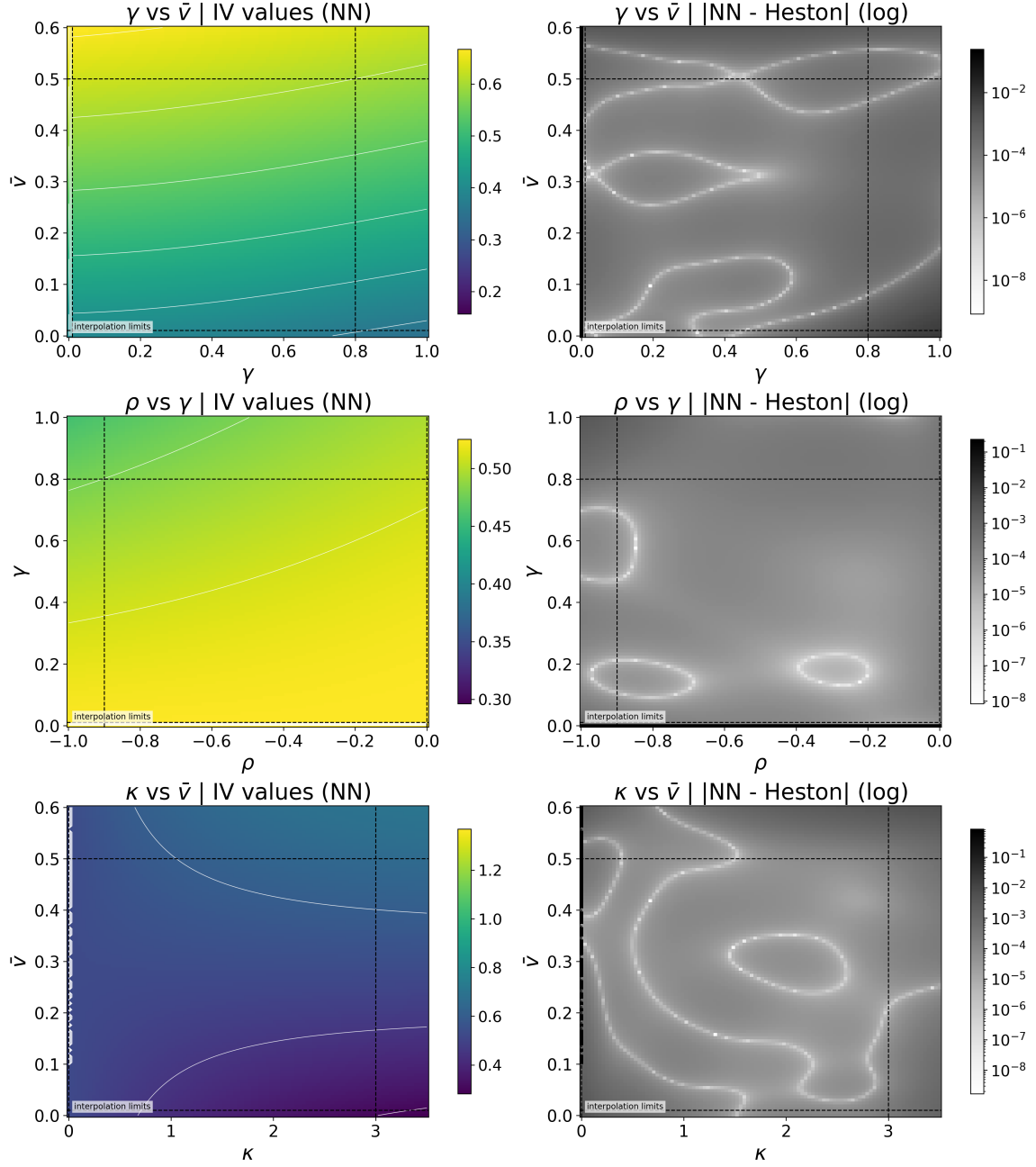


Fig. 6.1. Sensitivity grid for the ANN pricer. Top row: $(\gamma, \bar{v})$, middle row: (ρ, γ) , bottom row: $(\kappa, \bar{v})$. Left panels report IV-value maps with white isoclines; right panels report $\log |\text{NN} - \text{Heston}|$, where lighter tones indicate lower error and darker tones higher error. Dashed black lines indicate interpolation limits.

6.2.2. Regional Error Profile

Regional diagnostics identify where approximation error concentrates. On the test split, the largest MSE values are concentrated at low moneyness and short maturity: the bin $m \in [0.6, 0.8]$ accounts for roughly one quarter of test points, while $\tau \in [0.05, 0.25]$ accounts for roughly 6.8%. Interior and longer-horizon bins show a clear error drop, so the pattern is structural rather than driven by isolated rare points.

The same pattern appears in the regional MSE curves by moneyness and maturity bins. In both validation and test curves, the first bin is clearly the most difficult, while errors drop sharply and remain much lower in interior bins. This is consistent with boundary/extrapolation effects and with the smoother behavior observed around central bins.

6.3. Calibration Network Performance

This section reports the benchmark outcomes of the CaNN workflow. The compact calibration setup and diagnostic reading are introduced in Chapter 4; here we focus on final metrics and on the selection of the reference calibration run used in the rest of the chapter.

6.3.1. Benchmark Snapshot

The calibration benchmark in this section follows the inverse-problem construction of Chapter 4: it uses a fixed ANN surrogate (from Section 6.2) and one fixed quote universe of 35 contracts (Table 6.2). The calibrated vector is $(\rho, \kappa, \gamma, \bar{v}, v_0)$, optimized with Differential Evolution plus polish under the same bounds and controls defined in Chapter 4.

TABLE 6.2. QUOTE UNIVERSE USED IN THE CANN BENCHMARK.

Item	Value
Number of quotes	35
Grid structure	Complete 5×7 grid over (m, τ)
Moneyness nodes m	$\{0.85, 0.925, 1.00, 1.075, 1.15\}$
Maturity nodes τ	$\{0.50, 0.75, 1.00, 1.25, 1.50, 1.75, 2.00\}$
Interest rate r	0.03 (constant across quotes)

Source: Own elaboration

6.3.2. Fit Quality And Convergence

The final Liu-like calibration run uses the selected ANN width and depth from Chapter 3, with $\tanh$ hidden activations and Xavier uniform initialization. This $\tanh$ configuration supersedes the earlier ReLU calibration in the final benchmark because it gives a lower quote residual and a substantially better recovery of the synthetic truth parameters.

To characterize residual magnitudes directly in IV units, Table 6.3 summarizes the selected run with the same primary metric set used throughout the chapter.

TABLE 6.3. CANN QUOTE-LEVEL RESIDUAL METRICS
(SELECTED TANH RUN).

n	MSE	RMSE	MAPE (%)
35	7.43×10^{-10}	2.73×10^{-5}	0.0048

Source: Own elaboration

Residual concentration is very small at quote level. The largest absolute residual in the selected tanh run is 6.79×10^{-5} in IV units, and the signed residuals remain balanced around zero over the (m, τ) grid. The resulting quote fit is therefore one order of magnitude tighter than the earlier ReLU calibration used during the intermediate experiments.

6.3.3. Parameter Recovery Profile

Since this quote set includes sidocar synthetic truth, parameter-level errors can be reported directly. Table 6.4 keeps the comparison focused on the final selected run and the Liu et al. reference values.

TABLE 6.4. SELECTED TANH CANN RUN VERSUS LIU ET AL.
(ABSOLUTE PARAMETER ERRORS).

Metric	This work (selected tanh run)	Liu et al. [3]
Weighted MSE	7.43×10^{-10}	$\sim 10^{-8}$
$ \rho - \rho^* $	8.05×10^{-3}	4.84×10^{-2}
$ \kappa - \kappa^* $	8.48×10^{-4}	4.88×10^{-2}
$ \gamma - \gamma^* $	6.57×10^{-3}	3.28×10^{-2}
$ \bar{v} - \bar{v}^* $	1.02×10^{-4}	4.54×10^{-3}
$ v_0 - v_0^* $	1.83×10^{-4}	4.39×10^{-4}

Source: Own elaboration based on selected run outputs and [3].

In Table 6.4, *Weighted MSE* denotes the weighted mean-squared IV residual used as calibration objective. Under the replicated 35-quote Liu grid, the selected tanh run improves every reported parameter error relative to the Liu et al. values. The largest absolute deviations still occur in ρ and γ , but their scale is reduced from 10^{-2} – 10^{-1} to the 10^{-3} – 10^{-2} range. This is the strongest calibration result obtained in the thesis and is the CaNN result used in the final interpretation.

6.3.4. Local Smoothness And Curvature Study

To characterize local identifiability around the calibrated optimum, Jacobian and Hessian diagnostics were computed at θ^* for the selected tanh run.

TABLE 6.5. LOCAL-CURVATURE DIAGNOSTICS AT THE CALIBRATED SOLUTION θ^* (SELECTED TANH RUN).

$\ \nabla J(\theta^*)\ _2$	$\lambda_{\min}(H)$	$\lambda_{\max}(H)$	$\kappa_{\text{abs}}(H)$
8.75×10^{-6}	4.69×10^{-5}	4.18×10^1	8.89×10^5

Source: Own elaboration

The selected solution satisfies near-stationarity ($\|\nabla J(\theta^*)\|_2 \ll 10^{-4}$). The Hessian is symmetric, and the reported extreme eigenvalues are both non-negative in this run, consistent with a locally convex basin around the selected optimum. At the same time, $\kappa_{\text{abs}}(H) \approx 8.9 \times 10^5$ indicates pronounced anisotropy (flat and stiff directions coexist). The strongest cross-parameter coupling appears between $\bar{v}$ and v_0 (off-diagonal entry ≈ 18.47), followed by (γ, v_0) and $(\gamma, \bar{v})$, which is coherent with the observed recovery pattern.

The Jacobian components at θ^* remain close to zero (between 10^{-10} and 10^{-6}), reinforcing first-order convergence. The largest entries are associated with $\bar{v}$ and v_0 , but their magnitude is still very small, so there is no dominant residual slope left at the optimum.

The Hessian is the key second-order diagnostic. It has a block structure centered on $(\bar{v}, v_0)$: both diagonal entries are the largest (high local curvature), and their off-diagonal term is also large, which indicates strong local coupling between long-run variance and initial variance. By contrast, the κ -related diagonal curvature is much smaller, so perturbations in that direction are less strongly penalized around θ^* . This mixed pattern (stiff variance block plus flatter directions in other coordinates) explains why fit residuals can be very small while some parameters, especially ρ and γ , remain harder to pin down with the same precision as $(\bar{v}, v_0)$.

The results above define the selected CaNN benchmark used in the rest of this chapter.

6.4. PINN Pricing Performance

This section reports the baseline PINN pricing quality before introducing Sobolev terms in the training objective. The goal is to establish a reproducible reference for price-level accuracy and runtime against the COS/Heston engine.

The detailed reproducible setup (input convention, standardization, COS parameters, filtering, and arithmetic context) is moved to Appendix C. In short, the evaluated model is the baseline parametric PINN without Sobolev terms, benchmarked against the Heston-COS reference on 9.996×10^3 valid comparisons after numerical filtering.

6.4.1. Global Pricing Metrics

The compact global summary is embedded in Figure 6.2: $\text{MSE} = 3.1621 \times 10^{-5}$, $\text{RMSE} = 5.6233 \times 10^{-3}$, and $\text{MAPE} = 16.01\%$. These results are obtained with the baseline PINN only (without Sobolev augmentation). In runtime terms, the PINN evaluates the full benchmark in 0.0203 s ($\approx 4.91 \times 10^5$ points/s), while COS requires 114.47 s (≈ 87.36 points/s), i.e., an empirical throughput ratio of approximately 5.63×10^3 .

6.4.2. Spatial And Distributional Diagnostics

The pointwise agreement between PINN and COS prices, the global metric table, the spatial error maps, and the distributional diagnostic are consolidated in Figure 6.2. The scatter panel remains tightly concentrated around the identity line, while a mild negative bias (mean error -1.87×10^{-3}) indicates slight underpricing in the central price range.

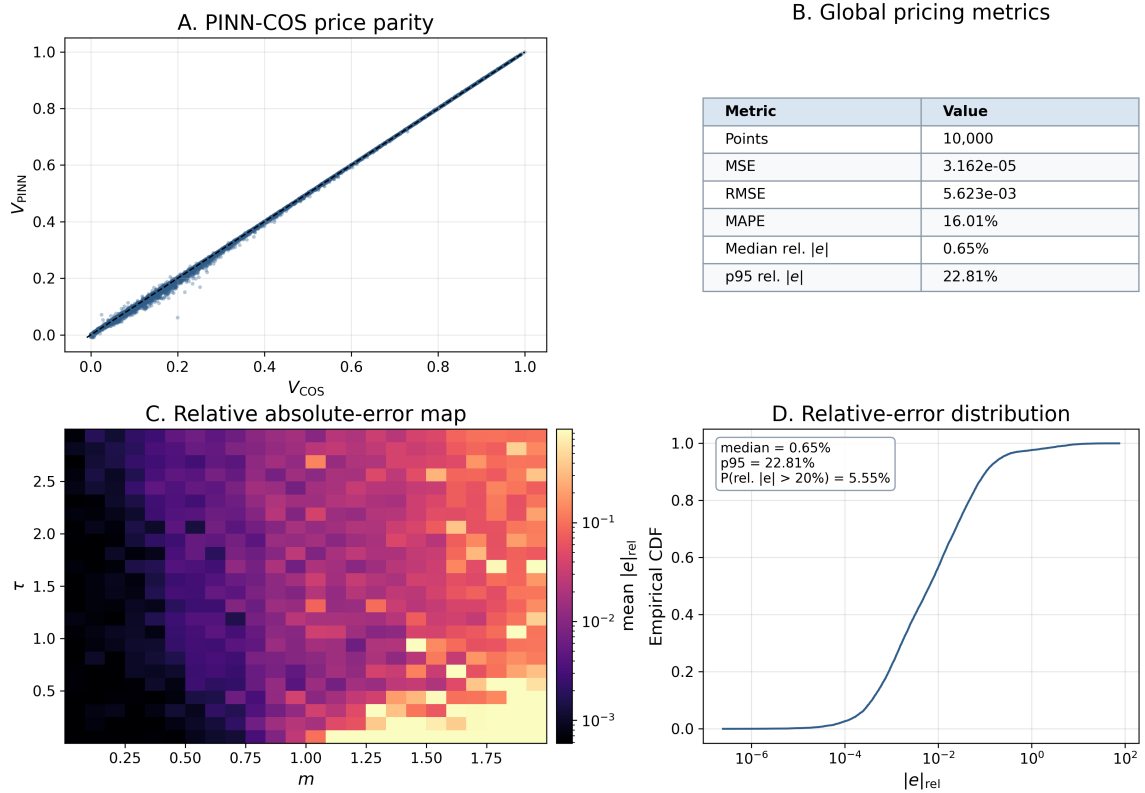


Fig. 6.2. Composite PINN pricing benchmark versus COS reference. Top-left: PINN-COS scatter. Top-right: global MSE/RMSE/MAPE and relative-error summary. Bottom-left: mean relative absolute-error map over (m, τ) . Bottom-right: empirical CDF of relative absolute pricing errors.

The spatial panel shows that relative error is not homogeneous over (m, τ) : the largest relative deviations are concentrated in low-reference-price regions, especially towards high moneyness and shorter maturities, whereas the central region is markedly more stable. These values must be interpreted jointly with the absolute-unit metrics (MSE and

RMSE): when the COS reference price is close to zero, relative error inflates, so local high percentages do not necessarily correspond to large monetary mispricing. Distributionally, the median relative absolute error is 0.65%, the p_{95} relative absolute error is 22.81%, and 5.55% of points exceed 20% relative error.

6.5. PINN Greeks Performance

This section evaluates autodiff-based Greek quality for the baseline PINN and quantifies the effect of Sobolev augmentation through a controlled ablation.

6.5.1. Ablation Design And Reproducible Protocol

The ablation isolates the impact of Jacobian-informed Sobolev terms on top of the baseline PINN objective, comparing:

- **Baseline PINN:** PDE + terminal + lower-boundary losses only.
- **Sobolev-augmented PINN:** baseline loss plus derivative supervision terms $\mathcal{L}_{g,1}$ and $\mathcal{L}_{g,2}$ as defined in Chapter 5.

To keep the comparison interpretable, both variants are benchmarked on the same (m, τ) grid, with identical fixed parameters, the same Heston characteristic-function Greek engine, and the same numerical integration settings.

TABLE 6.6. REPRODUCIBLE PROTOCOL FOR THE GREEK ABLATION BENCHMARK.

Item	Value
Compared variants	Baseline PINN vs Sobolev-augmented PINN
Architecture and features	Fixed across variants (same parametric PINN input convention)
Evaluation grid	$m \in [0.8, 1.2]$ (161 points), $\tau \in [0.05, 2.0]$ (161 points)
Total evaluation size	$n = 2.5921 \times 10^4$ points per Greek
Fixed model parameters	$\nu = 0.04$, $\rho = -0.7$, $\kappa = 2.0$, $\gamma = 0.3$, $\bar{\nu} = 0.04$, $r = 0.01$
Reference Greeks	Semi-analytical Heston characteristic-function Greeks
Characteristic function integration setup	$u_{\min} = 10^{-6}$, $u_{\max} = 200$, $n_u = 1200$
Shared conventions	Put option, strike $K = 1$, THETA sign convention ($\text{THETA} = -\partial V / \partial \tau$), MAPE floor 10^{-4}
Arithmetic context	Double precision on CPU

Source: Own elaboration

6.5.2. Ablation Metrics By Greek

Greek quality is reported with the same primary metrics used in the rest of the chapter: MSE, RMSE, and MAPE (%).

TABLE 6.7. GREEK METRICS FOR BASELINE AND SOBOLEV PINN ($N = 2.5921 \times 10^4$ PER GREEK).

Variant	Greek	MSE	RMSE	MAPE (%)
Baseline	DELTA	1.11×10^{-3}	3.33×10^{-2}	10.74
Baseline	GAMMA	1.48×10^{-1}	3.85×10^{-1}	82.98
Baseline	VEGA	4.30×10^{-3}	6.55×10^{-2}	71.84
Baseline	THETA	2.32×10^{-5}	4.82×10^{-3}	31.98
Baseline	RHO	7.01×10^{-3}	8.37×10^{-2}	95.00
Sobolev	DELTA	1.23×10^{-7}	3.51×10^{-4}	0.28
Sobolev	GAMMA	7.99×10^{-5}	8.94×10^{-3}	31.29
Sobolev	VEGA	1.00×10^{-5}	3.17×10^{-3}	36.11
Sobolev	THETA	2.81×10^{-7}	5.30×10^{-4}	1.88
Sobolev	RHO	2.05×10^{-6}	1.43×10^{-3}	9.05

Source: Own elaboration

Compared with the baseline PINN, Sobolev augmentation reduces RMSE by approximately $9\times$ (THETA) to $95\times$ (DELTA), with the largest practical gains in the previously hardest Greeks (GAMMA and VEGA). Greek-inference throughput decreases from 9.48×10^3 to 8.60×10^3 points/s ($\sim 9.3\%$ reduction), which remains a clearly favorable trade-off given the global accuracy gains.

6.5.3. Visual Diagnostics

This subsection is organized as a before/after narrative. Baseline diagnostics are first used to localize the dominant Greek-error modes, and Sobolev diagnostics are then read with the same visual layout to make improvements directly comparable.

Figure 6.3 displays baseline relative absolute-error maps for the five Greeks. The spatial pattern is heterogeneous, with broad high-error regions in GAMMA and VEGA, which is consistent with their weaker baseline MSE/RMSE values in Table 6.7.

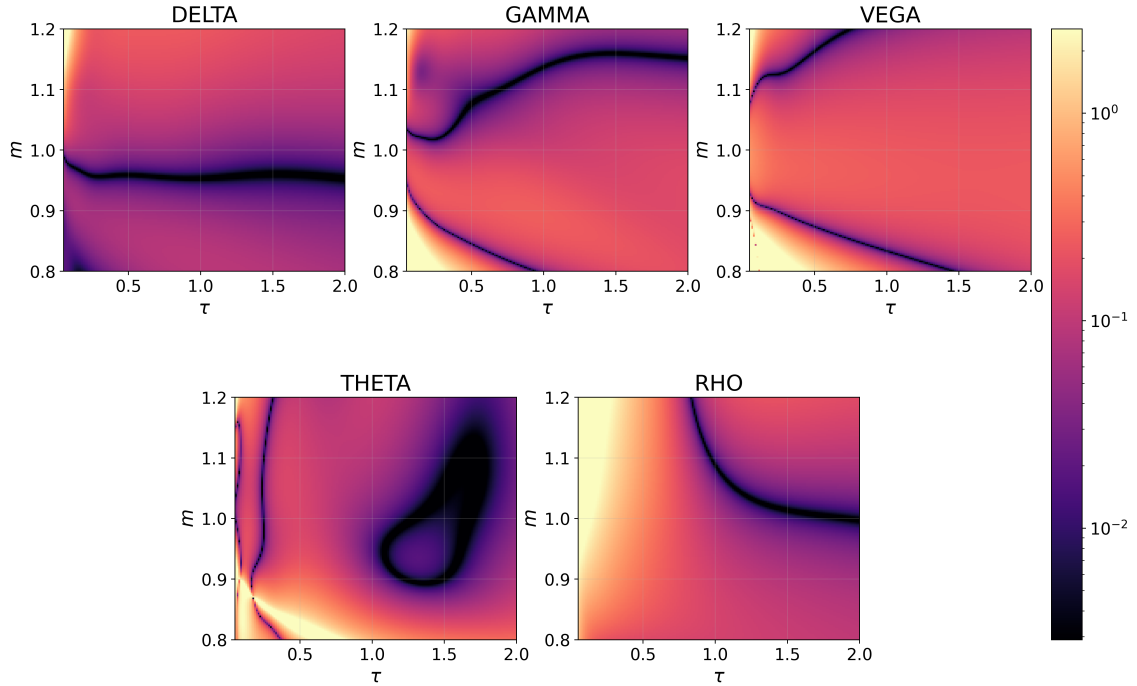


Fig. 6.3. Baseline PINN Greek relative-error maps.

The baseline histogram panel complements this map-level view with error distributions. The heaviest tails appear in GAMMA and VEGA, while DELTA, THETA, and RHO are comparatively tighter. This confirms that the baseline model captures coarse sensitivity structure but still exhibits non-negligible extreme deviations in the hardest derivatives.

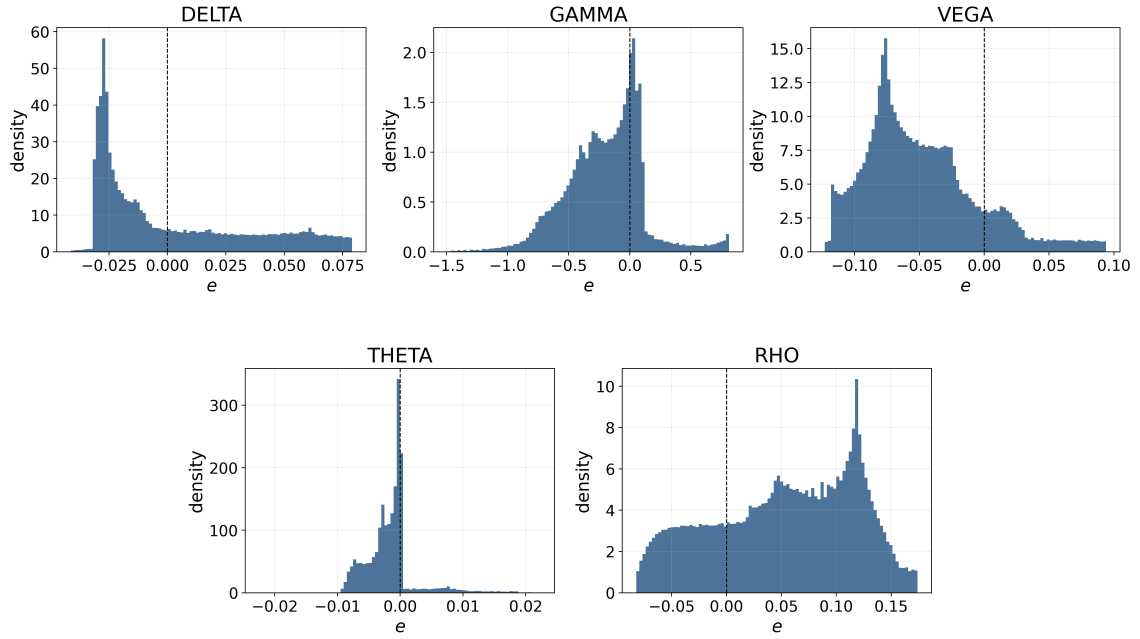


Fig. 6.4. Baseline PINN Greek error histograms.

After adding Sobolev terms, Figure 6.5 shows a clear contraction of high-error regions across all Greeks. The strongest visual improvement appears in GAMMA and VEGA, where problematic zones become much smaller and less intense than in the baseline maps. This spatial change is consistent with the order-of-magnitude RMSE reductions reported in Table 6.7.

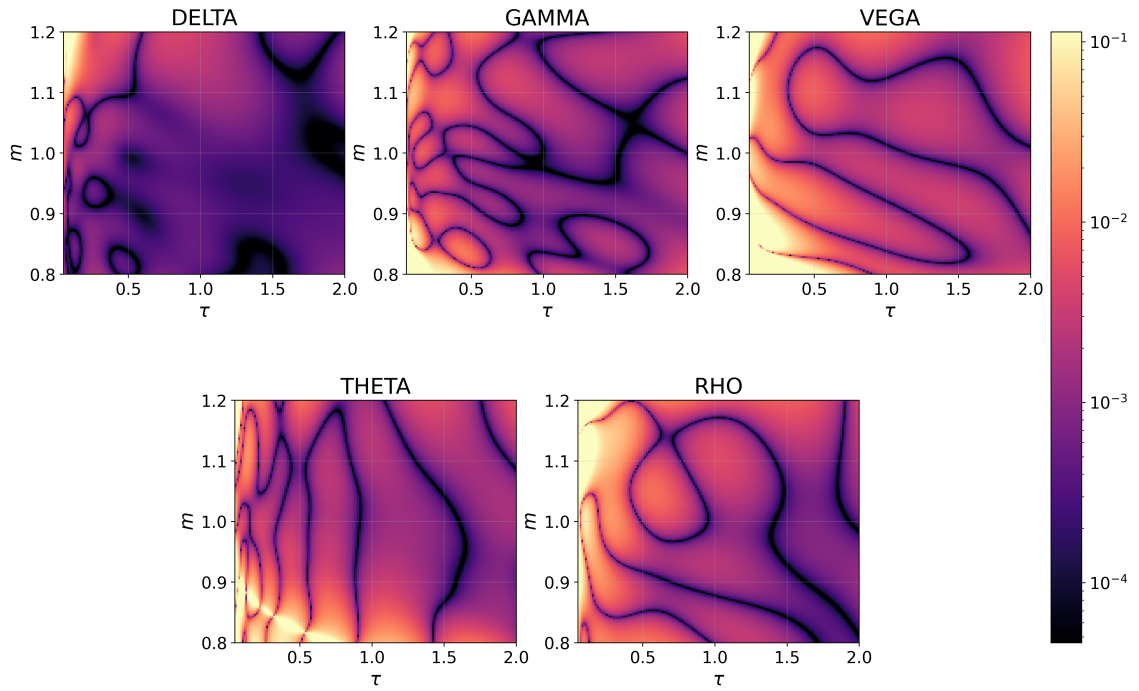


Fig. 6.5. Sobolev PINN Greek relative-error maps.

The same effect appears in Figure 6.6: Sobolev training narrows the distributions

and reduces tail mass, so large deviations become less frequent for all five Greeks compared with the baseline histogram panel. The tail compression is particularly visible in GAMMA and VEGA, where baseline outliers were most persistent.

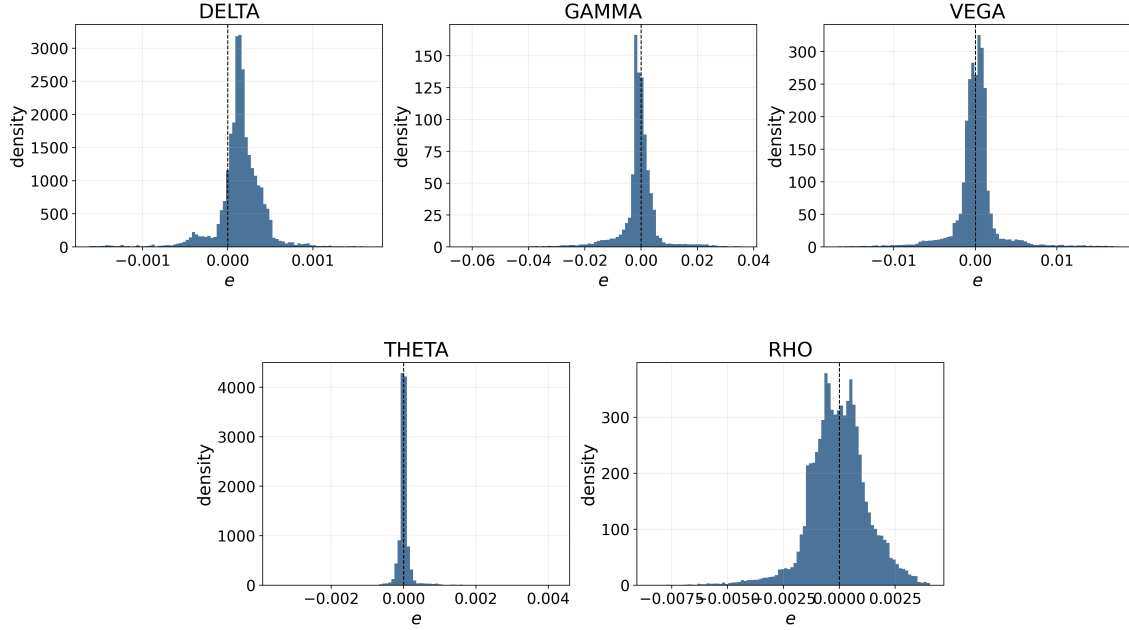


Fig. 6.6. Sobolev PINN Greek error histograms.

The two-phase optimization schedule used in this ablation trains epochs 0 to 4000 without Sobolev terms, then activates Sobolev augmentation from epoch 4000 onward. Together, the diagnostics indicate that Sobolev augmentation improves local derivative fidelity, not just aggregate scalar scores, while keeping the runtime overhead limited.

6.5.4. Control-Variate Greek Diagnostic

An additional Greek-focused diagnostic was run in parallel with the Sobolev study using an analytic-control-variate (ACV) hard patch. The accepted best-gate candidate keeps the frozen baseline PINN, adds a local Black–Scholes put control variate around the short-maturity ATM layer [5], and initializes the residual network at zero without residual training. Consequently, the result isolates the effect of the structural control variate rather than an extra supervised Greek fit.

In this diagnostic, “hard” refers to the short-maturity ATM region

$$\mathcal{D}_{\text{hard}} = \{(m, \tau) : |\log(m)| < 0.03, \tau < 0.05\},$$

which is precisely the neighborhood where the terminal put payoff kink at $m = 1, \tau = 0$ creates the strongest curvature stress for Gamma estimation.

TABLE 6.8. ACV CONTROL-VARIATE DIAGNOSTIC AGAINST THE ABLATION BASELINE.

Metric	Baseline PINN	ACV control variate
Price RMSE	4.269×10^{-3}	4.235×10^{-3}
Delta RMSE	3.381×10^{-2}	3.269×10^{-2}
Gamma RMSE	5.348×10^{-1}	4.040×10^{-1}
Hard Delta RMSE	8.729×10^{-2}	1.288×10^{-2}
Hard Gamma RMSE	4.992	6.446×10^{-1}
Hard p_{99} Gamma error	17.643	1.612

Source: Own elaboration based on ablation-study outputs.

The control variate is therefore the strongest candidate in the ablation table for full Delta RMSE, full Gamma RMSE, hard-region Delta RMSE, hard-region Gamma RMSE, and hard p_{99} Gamma error. The result should still be read cautiously near the terminal payoff kink: the remaining Gamma error is concentrated in the short-maturity ATM layer, and the extremely short bucket $\tau < 5 \times 10^{-4}$ still requires separate validation of the semi-analytical Gamma reference before making stronger claims there.

6.6. Cross-Model Computational Comparison

This section summarizes the end-to-end pipeline in operational order, emphasizing where each model contributes and how computational cost is distributed.

First, market calibration is driven by the ANN pricer plus the CaNN optimization loop. The ANN stage provides fast implied-volatility evaluations ($\approx 8.43 \times 10^5$ points/s), while CaNN concentrates cost in iterative inverse fitting. The selected tanh calibration reaches quote-level residual RMSE 2.7254×10^{-5} and improves all reported Liu et al. parameter errors on the replicated 35-quote benchmark.

Second, the calibrated market state $(\rho, \kappa, \gamma, \bar{v}, v_0)$ is injected into the parametric PINN pricing stage. In this configuration, baseline PINN pricing reaches MSE 3.16×10^{-5} , RMSE 5.62×10^{-3} , and MAPE 16.01%, while running at $\approx 4.91 \times 10^5$ points/s. Under the same benchmark protocol, COS runs at ≈ 87.36 points/s, so PINN remains several orders of magnitude faster in repeated evaluation mode.

Third, Greek extraction is performed on top of the same calibrated parameter state. The baseline PINN Greek pass runs at $\approx 9.48 \times 10^3$ points/s, while the Sobolev variant runs at $\approx 8.60 \times 10^3$ points/s (about 9.3% slower) and substantially improves derivative accuracy (for example, GAMMA RMSE from 3.85×10^{-1} to 8.94×10^{-3} , VEGA RMSE from 6.55×10^{-2} to 3.17×10^{-3}). In the separate hard-region ablation, the ACV control variate gives the best Delta/Gamma metrics, reducing hard Gamma RMSE from 4.992 to

6.446×10^{-1} and hard p_{99} Gamma error from 17.643 to 1.612.

Pipeline-level error should be read as stagewise bounds, not as a single scalar. The ANN and CaNN stages control IV-space mismatch used for calibration, pricing quality is validated in price space against COS, and Greek quality is validated in derivative space. Since these are different output spaces and objectives, the most reliable reading is the per-stage error envelope reported above and in Sections 6.2–6.5.

7. CONCLUSION AND FUTURE WORK

7.1. Main Conclusions

This thesis has developed and benchmarked an end-to-end computational workflow for Heston-based derivative analytics. The workflow starts from controlled data generation and numerical reference pricing, builds a supervised ANN implied-volatility surrogate, uses that surrogate inside a neural calibration loop, and then evaluates PINN-based pricing and Greek computation against semi-analytical references. The main outcome is not a single model, but an executable and auditable pipeline in which each stage can be tested independently and then connected to the next one.

The first conclusion is that the supervised ANN surrogate remains a useful acceleration layer for calibration. The ANN benchmark shows that the implied-volatility map can be approximated with errors in the 10^{-3} IV range on held-out data, while preserving very fast inference. This makes it suitable as a forward evaluator inside repeated inverse-problem calls, provided that the input remains inside the interpolation region and that boundary/extrapolation zones are treated with caution.

The strongest calibration result is obtained with the Liu-like tanh ANN used in the final CaNN benchmark. On the replicated 35-quote Liu grid [3], the selected run reaches weighted MSE 7.43×10^{-10} and residual RMSE 2.73×10^{-5} in IV units. More importantly, all reported Heston parameter errors improve relative to the Liu et al. reference values [3] under the same synthetic-truth comparison reported in Chapter 6. This is the clearest positive empirical result of the thesis: the calibration component does not merely reproduce the reference neural-calibration framework, but improves its parameter recovery in the benchmarked setting.

The second conclusion is that PINNs are effective as fast differentiable pricing surfaces, but their accuracy profile is more delicate than the ANN calibration component. The baseline PINN pricing benchmark reaches RMSE 5.62×10^{-3} against the COS reference [2] while giving a throughput several thousand times higher than the reference COS loop in repeated evaluation mode. However, the spatial diagnostics show that relative errors concentrate in low-price regions and near short maturities. This confirms that the PINN should be read as a fast differentiable approximation with identifiable failure modes, not as a universal replacement for the numerical solver.

The third conclusion concerns Greeks. Automatic differentiation makes Greeks available directly from the learned price surface, but accurate prices alone are not enough to guarantee accurate derivatives, especially for curvature quantities such as Gamma. Two strategies improved this part of the pipeline. Sobolev augmentation [4] substantially reduced global Greek errors by adding derivative information during training. In parallel,

the ACV control-variate diagnostic gave the strongest hard-region Delta and Gamma results without adding supervised Greek training to the residual patch. In the hard short-maturity ATM region, Gamma RMSE was reduced from 4.992 to 6.446×10^{-1} , and the hard p_{99} Gamma error from 17.643 to 1.612.

Taken together, these results support a practical division of roles. The ANN/CaNN block is the most mature part of the workflow and gives a strong calibration result. The PINN block is valuable as a differentiable pricing and sensitivity engine, but its derivative quality requires additional structure, either through Sobolev-style derivative supervision or through analytic control variates tailored to the singular region.

7.2. Contributions

The first contribution is a reproducible benchmark pipeline. All stages of the study are organized around explicit configurations, fixed data splits, numerical reference settings, and stored diagnostics. This matters because model comparisons in derivative pricing can be misleading if solver tolerances, data filters, or evaluation grids change silently between experiments.

The second contribution is the improved neural calibration result. The final tanh CaNN run preserves the basic Liu-style calibration idea [3], but obtains a stronger inverse fit on the benchmark quote set. The improvement appears both in quote-level IV residuals and in parameter recovery, which is the relevant quantity for a calibration workflow.

The third contribution is the PINN Greek analysis. The thesis does not stop at price RMSE; it evaluates first- and second-order sensitivities against semi-analytical Heston references and identifies where derivative errors concentrate. This is important because a pricing surrogate that is visually accurate can still be unusable for risk management if its local slopes and curvatures are unstable.

The fourth contribution is the hard-region ablation. The experiments show that several seemingly natural modifications, such as payoff-aware parametrizations or adaptive collocation, can improve one metric while degrading another. The accepted ACV control-variate result is therefore not just another model variant: it is the outcome of a targeted ablation process focused on the region where the payoff kink makes Gamma hardest to approximate.

7.3. Limitations

The main limitation is that the benchmark remains synthetic and Heston-specific. This is appropriate for controlled methodological evaluation, but it does not prove robustness on noisy market surfaces, sparse quotes, bid-ask spreads, dividends, or changing interest-rate conventions. The calibration result should therefore be read as a strong controlled

benchmark, not as a final production calibration study.

A second limitation is the dependence on vanilla European structure. The COS reference, the semi-analytical Greek engine, and the ACV correction are all built around a setting where strong analytical information is available. This is useful for validation, but portability to exotic derivatives or path-dependent contracts would require replacing parts of the reference and diagnostic stack.

A third limitation is the payoff kink itself. The ACV result reduces the short-maturity ATM Greek error substantially, but it does not remove the mathematical difficulty created by the non-smooth terminal payoff. In particular, the extremely short-maturity bucket $\tau < 5 \times 10^{-4}$ still requires careful validation of the reference Gamma before making stronger claims about that layer.

Finally, the thesis evaluates computational speed as a research benchmark rather than as a production latency target. The code is executable and scalable in structure, but it has not been optimized as a low-latency service or validated under portfolio-scale operational constraints.

7.4. Future Work

The first future direction is market-data validation. The CaNN result should be tested on real implied-volatility surfaces with realistic quote filtering, liquidity weights, and bid-ask uncertainty. This would show whether the improved synthetic calibration transfers to market conditions and whether the parameter recovery remains stable when the truth is no longer known.

The second direction is a hedging-oriented evaluation of Greeks. The present thesis verifies Greeks pointwise against reference values, which is the necessary first step. A natural continuation is to connect the Greek engine to a simple hedging backtest or risk-reporting workflow, measuring whether improved Delta and Gamma translate into more stable hedge rebalancing and P&L attribution.

The third direction is model and payoff extension. The executable structure of the pipeline makes it possible to test alternative stochastic-volatility models, local-volatility variants, or other payoff classes. For exotic derivatives, the main challenge will be to replace vanilla-specific analytical references with suitable numerical or simulation-based validation tools.

The fourth direction is further machine-learning improvement. Promising candidates include better loss balancing for PINNs, residual-gradient penalties in the spirit of gPINNs [8], adaptive sampling with stricter price-preservation criteria, and architectures designed specifically for stable second derivatives. The ablation results in this thesis suggest that such improvements should be accepted only if they improve hard-region Greek accuracy without degrading global pricing quality.

Overall, the thesis demonstrates that neural methods can be useful in derivative analytics when they are embedded in a disciplined numerical benchmark. The strongest evidence is the improved CaNN calibration result and the hard-region Greek improvement obtained with the ACV control variate. The remaining work is to move from controlled Heston benchmarks to noisier market and portfolio settings while preserving the same level of diagnostic discipline.

BIBLIOGRAPHY

- [1] S. L. Heston, “A Closed-Form Solution for Options with Stochastic Volatility with Applications to Bond and Currency Options,” en, *The Review of Financial Studies*, vol. 6, no. 2, pp. 327–343, 1993. doi: [10.1093/rfs/6.2.327](https://doi.org/10.1093/rfs/6.2.327).
- [2] F. Fang and C. W. Oosterlee, “A novel pricing method for european options based on fourier-cosine series expansions,” *SIAM Journal on Scientific Computing*, vol. 31, no. 2, pp. 826–848, 2009. doi: [10.1137/080718061](https://doi.org/10.1137/080718061).
- [3] S. Liu, A. Borovykh, L. A. Grzelak, and C. W. Oosterlee, “A neural network-based framework for financial model calibration,” en, *Journal of Mathematics in Industry*, vol. 9, no. 1, p. 9, Dec. 2019. doi: [10.1186/s13362-019-0066-7](https://doi.org/10.1186/s13362-019-0066-7). Accessed: Oct. 21, 2025. [Online]. Available: <https://mathematicsinindustry.springeropen.com/articles/10.1186/s13362-019-0066-7>.
- [4] W. M. Czarnecki, S. Osindero, M. Jaderberg, G. Swirszcz, and R. Pascanu, “Sobolev Training for Neural Networks,” in *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, 2017. [Online]. Available: <https://papers.nips.cc/paper/7015-sobolev-training-for-neural-networks>.
- [5] F. Black and M. Scholes, “The pricing of options and corporate liabilities,” *Journal of Political Economy*, vol. 81, no. 3, pp. 637–654, 1973. doi: [10.1086/260062](https://doi.org/10.1086/260062).
- [6] P. Carr and D. B. Madan, “Option valuation using the fast fourier transform,” *Journal of Computational Finance*, vol. 2, no. 4, pp. 61–73, 1999. doi: [10.21314/JCF.1999.043](https://doi.org/10.21314/JCF.1999.043).
- [7] D. Hainaut and A. Casas, “Option pricing in the heston model with physics inspired neural networks,” en, *Annals of Finance*, vol. 20, no. 3, pp. 353–376, 2024. doi: [10.1007/s10436-024-00452-7](https://doi.org/10.1007/s10436-024-00452-7). [Online]. Available: <https://doi.org/10.1007/s10436-024-00452-7>.
- [8] J. Yu, L. Lu, X. Meng, and G. E. Karniadakis, “Gradient-enhanced physics-informed neural networks for forward and inverse PDE problems,” *Computer Methods in Applied Mechanics and Engineering*, vol. 393, p. 114 823, 2022. doi: [10.1016/j.cma.2022.114823](https://doi.org/10.1016/j.cma.2022.114823). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S0045782522001438>.
- [9] S. Wang, Y. Teng, and P. Perdikaris, “Understanding and mitigating gradient flow pathologies in physics-informed neural networks,” *SIAM Journal on Scientific Computing*, vol. 43, no. 5, A3055–A3081, 2021.
- [10] S. Wang, X. Yu, and P. Perdikaris, “When and why PINNs fail to train: A neural tangent kernel perspective,” *Journal of Computational Physics*, vol. 449, p. 110 768, 2022. doi: [10.1016/j.jcp.2021.110768](https://doi.org/10.1016/j.jcp.2021.110768). [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S002199912100663X>.

- [11] J. F. Urbán, P. Stefanou, and J. A. Pons, “Unveiling the optimization process of physics informed neural networks: How accurate and competitive can PINNs be?” en, *Journal of Computational Physics*, vol. 523, p. 113 656, Feb. 2025. doi: [10.1016/j.jcp.2024.113656](https://doi.org/10.1016/j.jcp.2024.113656). Accessed: Mar. 19, 2026. [Online]. Available: <https://linkinghub.elsevier.com/retrieve/pii/S0021999124009045>.
- [12] B. Oksendal, *Stochastic Differential Equations: An Introduction with Applications*, 6th ed. Springer, 2003.
- [13] F. D. Rouah and S. L. Heston, *The Heston Model and its Extensions in Matlab and C#*. John Wiley & Sons, 2013.
- [14] W. H. Press, S. A. Teukolsky, W. T. Vetterling, and B. P. Flannery, *Numerical Recipes: The Art of Scientific Computing*, 3rd ed. Cambridge University Press, 2007.
- [15] M. D. McKay, R. J. Beckman, and W. J. Conover, “A Comparison of Three Methods for Selecting Values of Input Variables in the Analysis of Output From a Computer Code,” en, *Technometrics*, vol. 42, no. 1, pp. 55–61, Feb. 2000. doi: [10.1080/00401706.2000.10485979](https://doi.org/10.1080/00401706.2000.10485979). Accessed: Mar. 21, 2026. [Online]. Available: <http://www.tandfonline.com/doi/abs/10.1080/00401706.2000.10485979>.
- [16] G. Cybenko, “Approximation by superpositions of a sigmoidal function,” *Mathematics of Control, Signals and Systems*, vol. 2, no. 4, pp. 303–314, 1989. doi: [10.1007/BF02551274](https://doi.org/10.1007/BF02551274).
- [17] K. Hornik, “Approximation capabilities of multilayer feedforward networks,” *Neural Networks*, vol. 4, no. 2, pp. 251–257, 1991. doi: [10.1016/0893-6080\(91\)90009-T](https://doi.org/10.1016/0893-6080(91)90009-T).

A. ANALYSIS OF THE COS METHOD ERROR

This appendix complements the Fourier-COS method introduced in Section 2.6. Its purpose is only to make explicit how the COS reference error is controlled in the implementation, since the detailed pricing formula is already given in Chapter 2.

A.1. COS Error Control

Starting from the pricing integral in Equation (2.21), the COS method replaces the transition density by the truncated cosine expansion in Equations (2.19)–(2.20). Following Fang and Oosterlee [2], the numerical error can be read as the sum of three effects: truncating the integration domain to $[a, b]$, using only N cosine terms, and approximating the exact density coefficients by characteristic-function-based coefficients. In the shorter notation used in Equation (2.25),

$$|V - V_N^{\text{COS}}| \leq \varepsilon_{\text{trunc}}([a, b]) + \varepsilon_N.$$

For plain-vanilla European options, the payoff coefficients H_k in Equation (2.23) are available in closed form, so no separate payoff quadrature error is introduced.

The truncation interval is selected from cumulants of the log-price state. Using the cumulants obtained from Equation (2.8), the implementation follows the standard cumulant rule

$$[a, b] = \left[c_1 - L \sqrt{c_2 + \sqrt{c_4}}, c_1 + L \sqrt{c_2 + \sqrt{c_4}} \right],$$

which is the interval stated in Equation (2.24). Increasing L reduces probability-mass truncation error, but it also widens the interval and may require a larger N to preserve resolution. Therefore, L and N are treated jointly rather than as independent numerical knobs.

A.2. Reference Configuration

In the experiments, the reference configuration uses $N = 1500$ and $L = 50$, with fallback $L = 80$ in low-IV or otherwise unstable inversion cases. These settings are deliberately conservative because COS prices are used downstream as ANN labels, implied-volatility inversion inputs, and PINN benchmark references. The benchmark protocol in Section 6.1 therefore checks COS convergence with respect to (N, L) , put-call parity, and Black–Scholes consistency in near-constant-variance regimes before neural-model errors are interpreted.

B. CALIBRATION OF THE HESTON MODEL

TODO:

C. PINN PRICING BENCHMARK SETUP

The pricing benchmark compares a baseline parametric PINN against the COS/Heston reference under a common evaluation protocol and shared numerical conventions.

The evaluated model is the baseline parametric PINN without Sobolev terms. Inputs follow the 8-dimensional convention

$$[\tau, m, v, \rho, \kappa, \gamma, \bar{v}, r],$$

with affine standardization fitted on training statistics. Concretely, each feature x_j is transformed as

$$\tilde{x}_j = \frac{x_j - \mu_j^{\text{train}}}{\sigma_j^{\text{train}}},$$

where μ_j^{train} and σ_j^{train} are computed only on the training split and then reused unchanged in validation, test, and benchmark evaluation. This keeps features on comparable scales and stabilizes training and gradient computation, which is especially important in PINN settings; it also avoids data leakage from evaluation sets.

The evaluation sample contains 1.0×10^4 candidates, with 9.996×10^3 valid comparisons after numerical filtering. COS pricing is computed with $N = 1500$, $L = 50$, and fallback $L = 80$, under European put convention ($K = 1$) and no-arbitrage tolerance 10^{-8} . All reported metrics are computed in double precision on CPU.

D. GLOBAL NOTATION GLOSSARY

This appendix contains a global glossary of symbols used throughout the thesis. It is intended as a living reference and can be expanded as additional notation appears in later revisions.

TABLE D.1. GLOBAL NOTATION GLOSSARY: CORE PRICING
AND MODEL NOTATION

Symbol	Meaning
t, T, τ	Valuation time, maturity, and time to maturity ($\tau = T - t$).
S_t, K, m	Spot price, strike, and moneyness ($m = S_0/K$).
r	Constant risk-free interest rate.
$\mathbb{Q}$	Risk-neutral probability measure.
V_t	Arbitrage-free option value at time t .
$V(t_0, x)$	Option value at valuation time t_0 when the state variable is x .
$V_{BS}, V_{\text{target}}$	Black–Scholes option value and target option value used in IV inversion.
Ψ	Option payoff function at maturity.
X_t, x, y	State variable and its initial/terminal values, typically $x = X(t_0)$ and $y = X(T)$.
F_X, f_X	Cumulative distribution function and probability density function of X .
θ	Generic model-parameter vector.
μ	Generic drift under the original probability measure.
$W_t^{(S)}, W_t^{(\nu)}$	Brownian motions driving the spot and variance processes.
$\kappa, \bar{\nu}, \gamma, \rho, \nu_0$	Heston parameters: mean reversion, long-run variance, vol-of-vol, correlation, and initial variance. Later chapters also use $\bar{\nu}$ and ν_0 for the variance parameters.

Source: Author's elaboration

TABLE D.2. GLOBAL NOTATION GLOSSARY: TRANSFORM,
COS, AND IV NOTATION

Symbol	Meaning
$\phi_X(u)$	Characteristic function of random variable X .
$\mathcal{M}_X(u), \zeta_X(u), \zeta_k$	Moment generating function, cumulant characteristic function, and k -th cumulant.
u, u_k, i	Fourier variable, COS frequency $u_k = k\pi/(b - a)$, and imaginary unit.
$[a, b], L, N, k$	COS truncation interval, truncation scaling parameter, number of terms, and series index.
$\bar{F}_k, H_k, U_k$	COS density, payoff, and Heston-COS payoff coefficients.
$\hat{V}_T(u), V_N^{\text{COS}}$	Fourier transform of the maturity payoff and N -term COS price approximation.
φ_H, C, D, d, g	Heston characteristic function and its affine auxiliary functions.
χ_k, ψ_k	Closed-form payoff integrals used in COS coefficients.
σ_{imp}	Implied volatility obtained from numerical inversion.
$F(\sigma), \varepsilon_{\text{price}}, \varepsilon_{\sigma}, n_{\text{max}}$	IV root function, price tolerance, volatility tolerance, and maximum number of solver iterations.
$\hat{\sigma}_i, \sigma_i^{\text{ref}}$	Predicted and reference implied volatilities for evaluation point i .
$\text{MSE}_{IV}, \text{RMSE}_{IV}, \text{MAPE}_{IV}$	Mean squared error, root mean squared error, and mean absolute percentage error computed in implied-volatility units.
Π_j, Ψ_j	Heston call-pricing probability terms and their Fourier integrands.
$\Delta, \Gamma, \text{Vega}, \text{Rho}, \Theta$	Standard option Greeks: spot first derivative, spot second derivative, variance/volatility sensitivity, interest-rate sensitivity, and time sensitivity.

Source: Author's elaboration